



Research review paper

Advances and opportunities in process analytical technologies for viral vector manufacturing

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ABSTRACT

Viral vectors are an emerging, exciting class of biologics whose application in vaccines, oncology, and gene therapy has grown exponentially in recent years. Following first regulatory approval, this class of therapeutics has been vigorously pursued to treat monogenic disorders including orphan diseases, entering hundreds of new products into pipelines. Viral vector manufacturing supporting clinical efforts has spurred the introduction of a broad swath of analytical techniques dedicated to assessing the diverse and evolving panel of Critical Quality Attributes (CQAs) of these products. Herein, we provide an overview of the current state of analytics enabling measurement of CQAs such as capsid and vector identities, product titer, transduction efficiency, impurity clearance etc. We highlight orthogonal methods and discuss the advantages and limitations of these techniques while evaluating their adaptation as process analytical technologies. Finally, we identify gaps and propose opportunities in enabling existing technologies for real-time monitoring from hardware, software, and data analysis viewpoints for technology development within viral vector biomanufacturing.

1. Introduction

Viral vectors (VVs) sit at the heart of rebirth in modern medicine and biotechnology because of their potential as effective delivery agents for gene therapies targeting an increasing portfolio of rare, orphan diseases (AGT, 2023; Bulcha et al., 2021), fighting aggressive forms of cancer, as vaccine platforms to counter infectious diseases and gateway to engineer cell therapies. The identity of biomanufacturing is simultaneously seeking to (i) implement de novo engineering and production of novel therapeutic modalities such as VVs or mRNA-based therapies, and (ii) maintain a sustained accelerated development timeline, in which any biotherapeutic can be designed, clinically tested, manufactured, and delivered to patients in 12 months – following the return to regular production from pandemic action.

Viral vectors have become a mainstream tool for delivering genetic payloads to various target tissues (Bashor et al., 2022; Bulcha et al.,

2021), solidifying their role in gene therapy as demonstrated by the recent uptick in regulatory approvals and the rapidly growing landscape of clinical trials (Arabi et al., 2022; Zhao et al., 2022). Manufacturing these at scale is a complex and relatively nascent platform in comparison to that of antibody therapeutics (Escors and Breckpot, 2010; Manilla et al., 2005; Perry and Rayat, 2021), and has continued to receive attention since their introduction in 1965 (Atchison et al., 1965), with the concurrent rise in disease indications requiring genetic intervention (Maragkou, 2022). Our survey on the trends in Orphan diseases (Orphanet database (“Orphanet Database,” 2023)) of which at least 80% have monogenic mutations as the pathophysiology (Condò, 2022) revealed increasing use of cell and gene therapies (Fig. 1A, B). This coupled with an increasing patient pool needing genetic therapies facilitated using viral vectors (Nguengang Wakap et al., 2020; Young et al., 2022) highlights how we can expect their demand to increase. VV manufacturing processes are an area of active research development,

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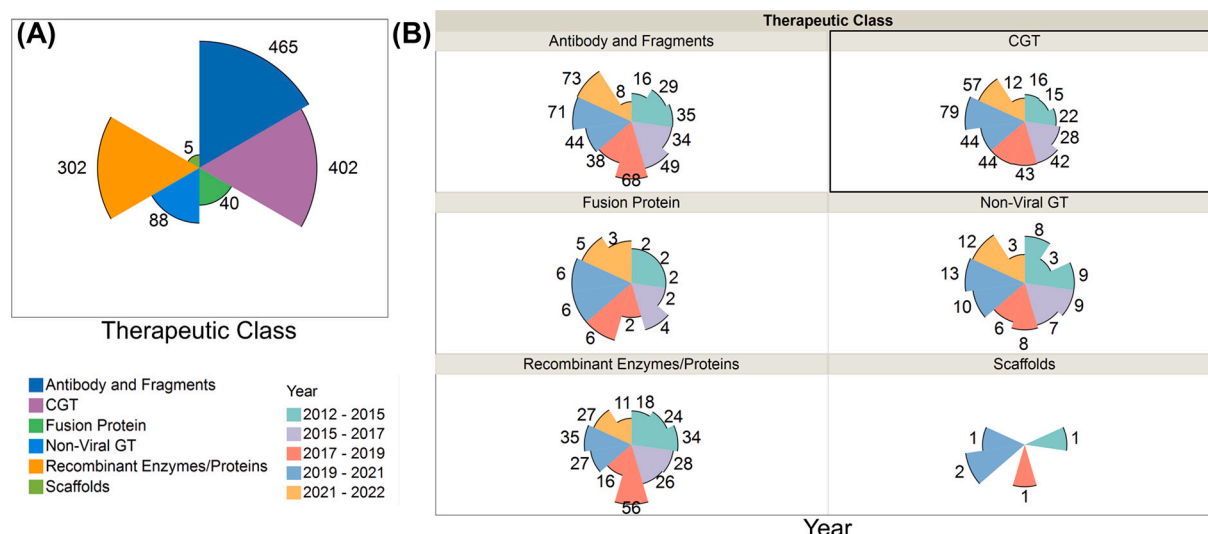


Fig. 1. (A) Landscape of therapeutics classes associated with orphan diseases including cell and gene therapies (CGT) and their indications surveyed in the last decade; (B) Classes of therapies indicated for orphan diseases between years 2012 and 2022 that shows increased perusal of cell and gene therapies (CGT) with the highest modal density in the last few years.

with challenges spanning across scalability, flexibility, and reproduction of product quality (Capra et al., 2022).

The potential of analytical technologies to be adapted across process scales is the measure of their ability to support flexible, robust, and affordable VV production (van der Loo and Wright, 2016; Vellinga et al., 2014). Accordingly, future VV manufacturing requires the introduction of flexible technologies capable of (i) supporting the production of diverse products, (ii) monitoring in real-time product titer and other CQAs across the production pipeline, and (iii) connecting seamlessly with a robust architecture for data acquisition and analysis and process control.

Adeno-associated viruses (AAV), Lentivirus (LV), Herpes simplex virus (HSV), and Adenovirus (AdV) top the non-exhaustive list of the current day vectors that are being researched the most for applications both in vivo and ex vivo (Pan et al., 2019). This choice is often based on the cell target for genetic modification and the ability of the VV to access it (tropism), pharmacology of the new gene (somatic vs. non-somatic cell integration), and their respective limits of genetic payload, following accurate analysis (Zhao et al., 2022). >400 of 964 registered gene therapy trials as of February 2023 (Chen, 2023) report the use of AAVs for in vivo indications – due to suitable tropism, and lower immunogenicity and genotoxicity (Grimm and Zolotukhin, 2015; Kotin, 2011; Rapti and Grimm, 2021). LVs have been explored in the contexts of stem cell engineering and CAR-T cell therapies that today compose the ‘ex vivo’ space (Lana and Strauss, 2020). VVs have also been indicated for treating infectious diseases like SARS-CoV2 such as AstraZeneca and Janssen’s pandemic vaccine that employed Ad26 vector (Sadoff et al., 2021) and more (enterovirus, rabies virus, other retrovirus (Draper and Heeney, 2010)) as vehicles for propagating oncolytic behavior or inducing absent protein.

Development of the existing VV analytical toolbox may have greatly benefitted from the plethora of knowledge available from the antibody manufacturing experience, especially initially with the identification and appointment of tools for assessing VV identity, purity, safety, and effectiveness as biotherapeutics. Many spectroscopic techniques developed like Raman, NIR, Fluorescence, optical reflectance and other inline chemo- and biosensors, have been extensively reviewed (Abu-Absi et al., 2014; Biechele et al., 2015; Challener, 2021; Fung Shek and Betenbaugh, 2021; Gargalo et al., 2022; Randek and Mandenius, 2018; Rathore, 2014; Reardon, 2021; Rolinger et al., 2020; Rosenberg and Cady, 2021; Teixeira et al., 2009; Tulsyan et al., 2020; Wasalathanthri et al., 2020), utilized for their benefits of enabling decision making such as nutrient

feeding, perfusion etc. (Framework, 2004; Gillespie et al., 2022) and seamlessly transferred to serve similar purposes within VV upstream manufacturing (Maruthamuthu et al., 2020; van der Loo and Wright, 2016). Key differences, however, involve the choice of cell lines used to produce VVs (HEK293, Vero, MDCK, HeLa, or Sf9), battery of product-related impurities that include (and are not limited to) empty particles, fragmented capsids, aggregated species, non-infectious vectors, variants with physical payload differences, unencapsulated genes of interest, isoforms with truncated capsids/pseudotypes, etc. (Schnödt and Büning, 2017; Wright, 2014), and necessary production additives such as transfection agents (such as residual plasmid DNA, chemical agents, or helper viruses), benzonase that become impurities at drug product stages, ultimately needing removal to ensure high therapeutic efficacy and avoid side effects like immunogenic reactions or hepatotoxicity, in patients receiving gene therapies (Leinonen et al., 2019; Troxell et al., 2023). Many of these attributes serve as critical quality attributes (CQA) (FDA, 2022) and are the primary targets for analytical technology development products (El Andari and Grimm, 2021; Escandell et al., 2022; Green and Lee, 2021). While their analysis is a necessity (Markova et al., 2022; Tustian and Bak, 2021; Wright, 2021), timely tracking of these attributes within manufacturing processes sets the stage for process analytical technologies (PAT) (CDER, 2004), and highlights the need for recognizing critical process parameters (CPPs) or measurable characteristics that directly influence quality attributes (Schad et al., 2023). As a general guidance, Table 1 supplies a general view of binned CQAs and Fig. 2(A) tracks and identifies typical CQAs and some previously associated classes of CPPs within a VV manufacturing process.

In this review, we discern the developments within the current state of analytics supporting VV manufacturing through the lens of suitability as PAT tools. A top-down approach to developing efficient PAT may start with the definition of their ‘line’ status based on type and need for monitoring, and is followed by the determination of single- vs. continuous-use, compatibility with process integration, durability, long-term stability, calibration, data logging and management, reusability vs. disposability, scalability, miniaturization, and portability etc. The evaluation culminates in the choice of appropriate materials, sensing elements, data processing mechanisms, and connectivity for process control, including wireless transmission (Gargalo et al., 2020). As the first line of definition, speed of analysis, physical distance from the unit operation, and connection to the process help categorize analytical technologies as in-line (analysis in situ), on-line (analysis at the process

Table 1

Categorized view of various quality attributes (non-exhaustive) for viral vector manufacturing, reported orthogonal analytical methods that provide similar information, with notes on their PAT readiness based on sample preparation needs, reported analysis volumes, time and generated data types to highlight the possibility of big data analysis and learning. Other attributes for safety testing (including ‡) have been excluded from the scope of this review.

Class	Attribute (Specific Markers)	Gold Standard Technique (s)	Orthogonal Technique(s)	Sample Preparation Needs (Orthogonal Tech.)	Sample Volume and Time (Orthogonal Tech.)	Notes on Data Type
Identity	Capsid Identity (also incl. Post Translational Modifications, Charge Variants, Amino Acid Composition)	Mass Spectrometry (CD-MS, RP-TOF-MS etc.)	Differential Scanning Fluorimetry	None	~50 µL Sample Volume, ~30 min Analysis Time	Discrete, Numerical
		Western Blot	Molecular signatures of VPs using Raman / FTIR Spectroscopy	None	Online/Real-time	Discrete and Continuous, Numerical
			CE-SDS for variations in VP populations		Offline / Days	Discrete and Continuous, Numerical
			HILIC + MS	MS Sample Preparation		MS
	Genome Identity (also incl. Insert Composition, Insert Length, SNV)	Next-generation sequencing			Offline / Days	
		CD-MS	Dot-blot hybridization (Complementality)		Offline / Days	Discrete, Numerical
		PCR based techniques + Gel Electrophoresis	Nanopore TRPS	Electrolyte Addition	~35 µL sample, 10 min acquisition	
	Pseudotype / Surface Markers	Western Blot	Multi-wavelength SV-AUC Immunoblotting	Dispersion Medium Addition	Offline / Days	Discrete, Numerical
			Flow Virometry		Offline / Days	Discrete, Numerical
	Strain Identification‡	Restriction Endonuclease Analysis	Mass Spectrometry	MS Sample Preparation	10 min Analysis Time	MS
Purity	Capsid Content, Concentration, Distribution	Capsid ELISA	SEC-MALS	None	~500 µL Sample Volume, ~30 min Analysis Time	Discrete and Continuous, Numerical
		cIEF	CE-SDS-LIF			Discrete and Continuous, Numerical
		SDS-PAGE	Nanoparticle Tracking Analysis	None	~500 µL Sample Volume, 1–5 min Analysis Time	Discrete and Continuous, Numerical
	Particle Integrity: Full versus Empty Particles, MW	AUC	(Cryo-)TEM	Microscopy (Fixing, Staining)	Offline / Days	Discrete, Numerical + Image
		AEX-HPLC	DGE-AUC	Dispersion Medium Addition	~500 µL Sample Volume, 10 min Analysis Time	Discrete, Numerical
		ddPCR + Capsid ELISA	Mass Spectrometry	MS Sample Preparation		MS
			FFF/SEC-MALS	None	~500 µL Sample Volume, ~30 min Analysis Time	Discrete and Continuous, Numerical
			Mass Photometry	None	~10–15 µL Sample Volume, 30 min Analysis Time	Discrete, Numerical
	GOI Packaging, Truncated Isoforms, Encapsidated Plasmid or Host DNA	AUC	SV-AUC	Dispersion Medium Addition	Offline / Days	Discrete, Numerical
		Western Blot	FFF/SEC-MALS	Centrifugation/Clarification depending on sample	~20–100 µL Sample Volume, ~25–60 min Run Time	Discrete and Continuous, Numerical
		ELISA	Next-generation sequencing		Offline / Days	
		PCR based techniques	CE + Staining (Molecular Beacons) + nES GEMMA	Dilution/Staining/Reagent Addition and MS Sample Preparation	Offline / Days	MS
			CD-MS	MS Sample Preparation	Offline / Days	MS
		SEC-MALS	Dynamic Light Scattering	None	Online/Real-time	Discrete and Continuous, Numerical
			Atomic Force Microscopy		Offline / Days	Discrete + Images
Polydispersity (Aggregates or Fragments, Particle Information)			Full-field Interferometry			Discrete, Numerical
			Flow Virometry	Microscopy (Fixing, Staining)		Discrete + Images
			(Cryo-)TEM	Microscopy (Fixing, Staining)	Offline / Days	

(continued on next page)

Table 1 (continued)

Class	Attribute (Specific Markers)	Gold Standard Technique (s)	Orthogonal Technique(s)	Sample Preparation Needs (Orthogonal Tech.)	Sample Volume and Time (Orthogonal Tech.)	Notes on Data Type
A	Helper Viruses		FFF/SEC-MALS based on MW	None	None, ~500 µL Sample Volume, ~30 min Analysis Time	Discrete and Continuous, Numerical
			Size or Charge-based Separation + DSC		~400 µL for separation and DSC, ~1 h based on resolution of temperature change	Discrete, Numerical
			Western Blot	Dilution/Staining/Reagent Addition	Offline / Days	
	hDNA	RT-PCR				
	HCPs	Host-specific HCP ELISA	LC-MS/MS (DDA, DIA, SWATH-MS)	MS Sample Preparation	Offline / Days	MS
	Process Residuals (Leachables, Benzonase, PEI, additives etc.)	Analyte-specific ELISA				
	Vector Genome Concentration	RT-PCR / qPCR	ddPCR	Dilution/Staining/Reagent Addition	Dye etc. addition, ~20 µL total sample x 96 samples, <50 mins	Discrete, Numerical
	Infectivity	Cell-based assays	Nucleic acid dye staining		Offline / Days	
		Analyte-specific ELISA	Flow Virometry	Microscopy (Fixing, Staining)	Dilution, optional staining, adequate sample and time required to record 100–1000 events	Discrete Numerical + Histograms + Images
	Potency	Next-generation sequencing	Plaque Assay		Offline / Days	Discrete, Numerical
			TEM	Microscopy (Fixing, Staining)	Offline / Days	Discrete + Images
			TRPS / Full-field Interferometry	Electrolyte Addition	Electrolyte addition, ~35 µL sample, 10 min acquisition	Discrete, Numerical + Histograms
	Transduction Efficiency	Cell-based assays				Discrete + Images
	Post-Transduction Functional Activity	TCID50	Cell-based / in vivo assay	Dilution/Staining/Reagent Addition	Offline / Days	Discrete + Images
		Western Blot	Enzymatic potency assays	Dilution/Staining/Reagent Addition	Offline / Days	Discrete, Numerical
	Replication Competence	RT-PCR	Plaque Assay	Dilution/Staining/Reagent Addition	Offline / Days	Discrete + Images
			Insertion site analysis		Offline / Days	Discrete
Safety‡	Stability	Native CD-MS	Differential Scanning Fluorimetry	Centrifugation/Concentration if needed	Addition of dyes, centrifugation or concentration; ~45 µL sample, PCR amplification time	Discrete, Numerical
			Variable Temperature CD-MS	Centrifugation/Concentration if needed	Concentration (if needed), ~25 µL sample, 8–16 min signal acquisition pre-processing	MS
			Electrophoretic Light Scattering	Centrifugation/Concentration if needed	Concentration (if needed), ~700 µL sample, <10 min signal acquisition	Discrete and Continuous, Numerical

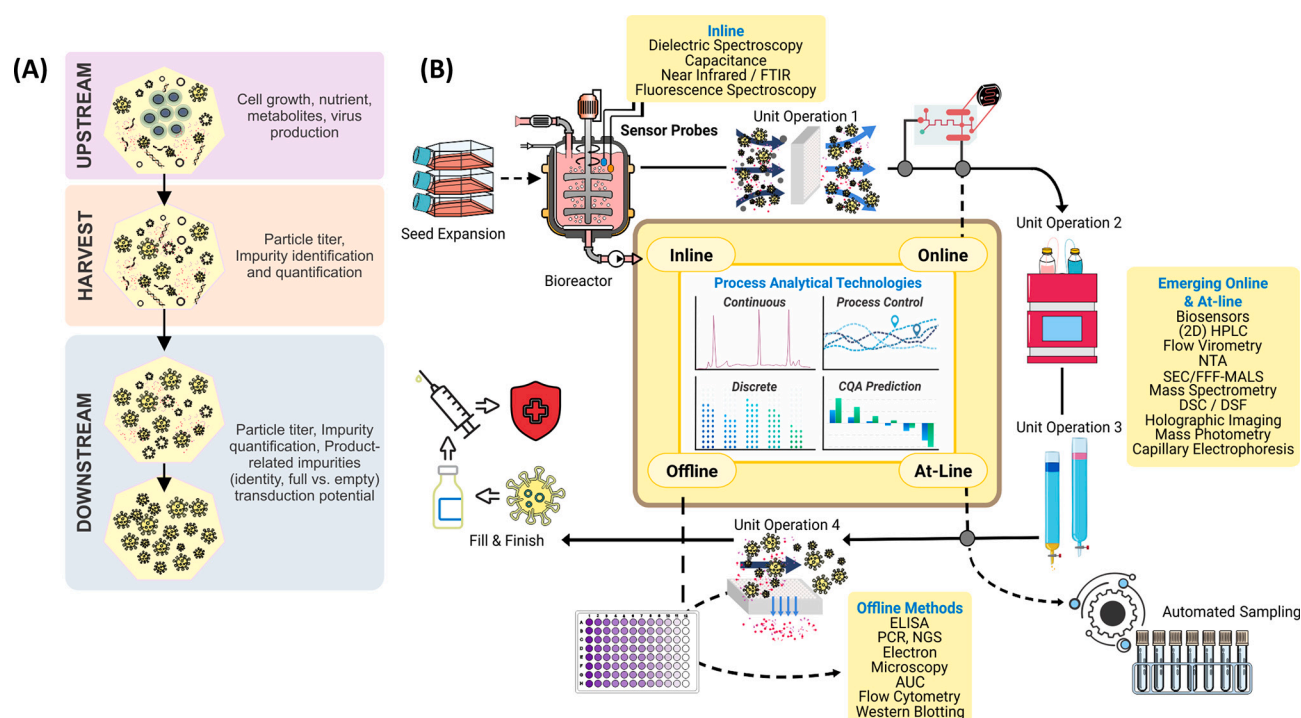


Fig. 2. (A) Schematic of typical CQAs and typically significant process parameters from upstream to downstream within a VV Biomanufacturing process, (B) Overview of Implementing Process Analytical Technologies within Viral Vector Manufacturing, with a perspective on currently utilized analytical methods as inline, online, at-line and offline technologies. A typical process and product progression of present day biomanufacturing [26] for viral vector-based gene therapies is also shown, with typical analytical information that can be obtained from each category of analytics.

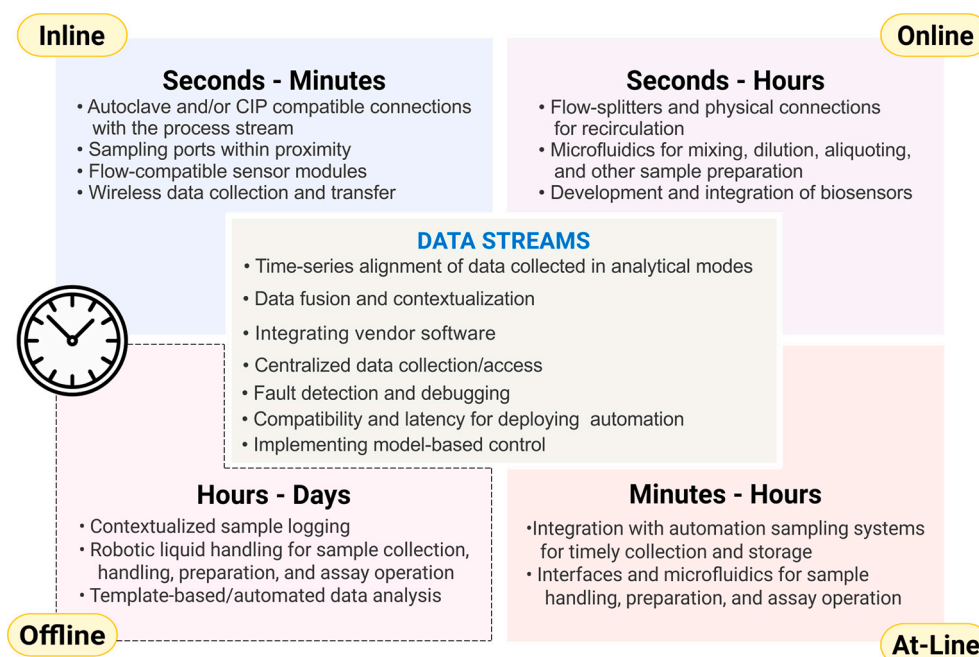


Fig. 3. Workflows of Inline, Online, At-line Process Analytical Technologies and general considerations for emerging technologies within these groups. Samples from a unit operation are typically directly transported through designed manifolds such as cell-free sampling systems or through recirculating streams into analytical instrumentation in the case of Inline/Online measurements, and intermediate steps such as the use of an automated sampling system followed by direct injection into cartridges, or the instrument is seen with at-line applications. While offline applications (within dotted lines) do not offer the benefits of real-time analytics, streamlining sample handling, processing, assaying, and data handling are avenues for advancement of such analytics.

site), at-line (analysis close to the process) and offline (Fig. 2(B)). Analytical characterization of the biopharmaceutical products is performed across the successive stages of processing, although only the

PATs that monitor CPP and CQA directly may be deployed on the manufacturing floor. Here, we survey the reported technologies that fulfill both requirements of CQA measurement and real-time or

continuous measurement, and that currently apply to various stages of process development. For each, we highlighted their advantages and identified needs for improvement at the respective stages of the process prior to discussing process control.

Given the diversity, biophysical complexity, and lability of VVs, sizable efforts have been devoted towards developing analytical assays and instrumentation across formats with considerations for method robustness, repeatability, sample preparation and real-time analysis/monitoring as avenues for development (Rathore and Sarin, 2023). These advancements fuel the movement towards the Industry 4.0/5.0 paradigms that call for (biomanufacturing) process evolution through digitization and integration of advanced analytics (Rathore and Sarin, 2023). We thus delve into analytics utilized within AAV, LV, and AdV manufacturing categorized by their readiness for real-time monitoring, feature available and orthogonal analytics by attribute (Table 1), and identify transformative opportunities for rapid analyses using microfluidics, sensors, or as multi-attribute monitoring (Fig. 3). We conclude with discussions on soft tools available for effective data comprehension and analysis including machine learning approaches that can help derive meaningful process insights, enable continuous, deeper learning, and potentiate process control, thus making PAT a reality within the VV manufacturing space. Collectively, we hope this review provides comprehensive information about the modern toolbox for monitoring VV CQAs – ultimately supporting clinical efforts and serving as a guide to researchers and engineers with various levels of expertise on the opportunities for further analytical technology development in this space.

2. Conventional offline analytics in viral vector manufacturing

Within the current manufacturing landscape, VV product characterization is performed *post facto*, promptly, based on process stage and/or production timetable. These assays comprise of often well-known, understood and widely implemented methods for estimating identity, titer, potency, purity, and safety (Grieger et al., 2006; Roldão et al., 2011; Wright, 2021). The standard toolbox typically involves analytical assays that are predominantly ‘offline’ (Section 2.1) which have already been extensively reviewed with notes on their robustness, throughput, associated costs and limitations (Altaras et al., 2005; Emerson and Glassey, 2021; Gimpel et al., 2021; Green and Lee, 2021; Hickey et al., 2023; Perry and Rayat, 2021). Table 1 adds to this knowledge by highlighting noted orthogonal techniques, and features enabling their use for real-time analysis. In this section, we also discuss traditional yet integral methods for VV process characterization and the effect of incorporating strategies like automation etc. to improve efficiency of their use.

2.1. Standard viral vector characterization toolbox: enzyme linked immunosorbent assay (ELISA), gel electrophoresis (SDS-PAGE, Western blotting), polymerase chain reaction (PCR), flow cytometry, next generation sequencing

Lothert et al. (2022) recently published a review of traditional approaches used for assessing virus particle infectivity (TCID₅₀, Titration via Flow Cytometry or Plaque formation, indirect methods such as Laser Force Cytology etc.), antigenic content (antigen specific ELISA, Hemagglutination, Single Radial Immunodiffusion etc.), methods for the quantification of genetic payload (qPCR, ddPCR) and particle counts (indirect ELISA, differential centrifugal sedimentation) as used within VV manufacturing. SDS-PAGE is also ubiquitously used for identifying various species via their migration and molecular weights. Though predominantly offline, it is routinely used for qualitative identification of viral proteins and thus indirectly particles (Lothert et al., 2022). Subsequent western blotting has also enabled assessing the stage of virus maturation like in the case of LV (Hyseni et al., 2020; Ruscic et al., 2019) and is often considered a gold standard assay for the presence and

identity of virus particles (Martins et al., 2001). ELISA and ddPCR amongst these are plate-based/multi-array assays that are capable of increased analysis throughput.

ELISAs once qualified for their sensitivity and robustness are ubiquitously used for the detection and quantification of various single-analytes such as capsid proteins or other surface antigens or multiple analytes, like host cell proteins. However, they are expensive tests, typically require triplicate measurements, have reports of inter-assay variability and often present with limited reliability due to high coefficient of variance, as high as 25% (Aucoin et al., 2008), and low robustness driven by consumable inconsistency or operator/operation errors. Along with the already imposed need for in-house assay development to cater to a specific analyte, this highlights the need for advances in molecular recognition and device technologies that can reduce variations along with improved, sensitive detections, that can potentially also be used with a variety of analyte – receptor combinations. Many advances in ELISA development have been reported for AAVs including development of universal receptors that can detect all AAV serotypes, creating *virelisa* (Cui et al., 2019), but questions about cross-reactivity due to the significant similarities between serotype capsid proteins still linger (Serrano et al., 2023). Furthermore, the recent automation of ELISAs (e.g., Ella Automated Immunoassay System etc.) and their coupling with automated sample-drawing and liquid-handling robots (e.g., Tecan or Hamilton) may warrant their transition from an offline to an at-line PAT (Anderson et al., 2022; Silva et al., 2021). In this context, integrating automated ELISAs with rapid data collection and analysis tools promises to reduce operational errors and open doors for ELISA-based multi-analyte detection. These transformations are positioning ELISAs to serve as PATs in processes operating at high characteristic time regimes – i.e., where unit operations feature large volume and residence times, such as viral vaccine manufacturing – where automated ELISAs and data-processing models can enable real-time decisions. For example, samples from various process points can be collected using automated sampling systems in case of continuous processing, and their manual delivery to liquid handling systems expedites sample preparation and subsequent analysis. This information could therefore also be retrieved quickly to make essential decisions. Examples of such at-line analytics implementation have been discussed in Section 4.3.

Knowledge of nucleic acid cargo (gene of interest, GOI) in terms of sequence identity, and GOI encapsulated VV concentration and purity is quintessential for making key decisions on safety and dosage. Quantitative PCR (qPCR) has historically been used to analyze the biochemically different payloads across various VVs (Barczak et al., 2015; Gimpel et al., 2021; Lothert et al., 2022; Sena-Esteves and Gao, 2018; Shmidt and Egorova, 2022; Vamva et al., 2020), and factors such as primer design and amplification conditions have been identified as critical for improving the accuracy of this method (Brimble et al., 2023). Droplet digital PCR (or ddPCR) incorporates microfluidic approaches to fluidize single DNA droplets even from crude samples (Duong et al., 2023; Lothert et al., 2022), thus providing an improved platform for transgene analytics while also eliminating the need for standard-based calibration. Furthermore, ddPCR systems such as Bio-Rad’s QX200 provide a highly automated, high throughput sample processing and analysis system which only requires manual intervention for plate transfers between droplet generation, amplification, and detection stages. It has also been noted that ddPCR in comparison to traditional qPCR methods increases assay robustness, with notably lower, ~10% difference in coefficients of variance (Dobnik et al., 2019). Other commercially available instruments for sample handling, extraction, and preparation of nucleic acids including cell-free, are also available (e.g., KingFisher Flex). Novel strategies for the development of multiplexed assays combining PCR and enzyme hybridization have also been noted in the case of AdV (Karen et al., 2004) and with 2D-ddPCR informing structural integrity in AAVs (Furuta-Hanawa et al., 2019). It is to be noted that free nucleic acids within VV manufacturing streams are considered process related impurities, which may be assayed using intercalating dyes such as

PicoGreen when sourced from the producer cell line. However, recent reports also compile the occurrence of host DNA packaged within these VVs or even chimeric complexes (Brimble et al., 2023), whose mechanisms and distributions are yet to be understood to devise efficient analytics. PCR also enables testing for undesirable replication competence of VVs, as developed for LV (Skrdlant et al., 2018) and to detect adventitious viruses (Barone et al., 2020). Limitations of the PCR technique for genomic quantification include bias from empty/full particle ratios measuring up to the same physiological genomic content (Schmidt and Egorova, 2022), or differentiating product isoforms at relatively lower concentrations without appropriate primers (Brimble et al., 2023) – to overcome which, next generation sequencing (NGS, Illumina, Nanopore etc.) has been used. Though offline and time consuming, bottom-up NGS offers unparalleled sensitivity in detecting nucleic acid impurities that may arise from various sources (Brimble et al., 2023), and thus vector genome identities. These analyses can be performed with massive parallelization and within timespans of days but need intervention and expertise in bioinformatics for data interpretation (Guerin et al., 2020; Lecomte et al., 2015; Maynard et al., 2019; Troxell et al., 2023).

Transduction assays are predominantly offline yet indispensable, and the most clinically relevant analytical technique that enables measuring the potency or efficacy of VVs in bringing about desired therapeutic effects. Techniques such as TCID₅₀, Plaque formation assays and Flow Cytometry, that have been reviewed for being the most used for this purpose (Lothert et al., 2022), however remain offline due to the involvement of cell growth, infection and hence considerable sample preparation, which are rate- and time-limiting factors. Recent developments in automation for sample preparation, image processing-analysis-readout (Hochdorfer et al., 2022) and using simpler instrumentation for detecting successful infection events (Labisch et al., 2021) have been reported, that can increase efficiency and accuracy of running these tests, albeit in offline mode, keeping economic constraints in mind. Limitations in the form of decision-related or robotic automation may arise however, encompassing qualification or validation challenges within the former and operational errors in the latter, requiring frequent calibration and maintenance.

2.2. Analytical ultracentrifugation

Analytical Ultracentrifugation (AUC) helps monitor macromolecule separation under sedimentation (Cole et al., 2008; Edwards et al., 2020) which is governed by their hydrodynamic radius, size, and shape – which for VVs has translated to separating empty, full, and partially fragmented genome particles based on density differences. Cesium Chloride (CsCl) and sucrose are extensively used as dispersing medium to fractionate samples, and sedimentation velocity profiles depend on the CsCl concentrations or type of viral vector (sucrose for LV), thereby needing high consistency and control in sample preparation and execution for repeatable results (Olgun et al., 2019; Yang et al., 2008).

AUC methods for various VVs including AdV (Yang et al., 2008) and AAVs (de la Maza and Carter, 1980) are available, with welcome improvements like that by Maza et al., who combined density centrifugation and velocity sedimentation giving rise to purification-ready analytical ultracentrifugation method for AAV characterization utilizing CsCl banding (Hoggan et al., 1966) or density gradient equilibrium AUC (DGE-AUC), that is used for efficient high-throughput AdV packaging analysis (Sternisha et al., 2023) and a capability of processing up to 21 samples within a span of 80 min. Purity and DNA content of viral capsids belonging to five AAV serotypes (AAV2, AAV5, AAV6, AAV8, and AAV9) have also been studied using sedimentation velocity-based AUC (SV-AUC) only to realize their high resolution in analyzing AAVs by quantifying empty, filled, and partially filled capsids – compared to traditional orthogonal methods like analytical chromatography (AEX), TEM, or mass photometry (Richter et al., 2023; Sternisha et al., 2023), while also establishing specific absorbance ratios, response factors, and

extinction coefficients for each capsid type. Furthermore, SV-AUC has been utilized to gain fundamental understanding of the relationship between smaller size and higher rep protein concentration which enhances packaging efficiency in an AAV triple transfection process (Burnham et al., 2015) and also provides the flexibility to use various detection mechanisms including optical, fluorimetric (Cole et al., 2008), refractometry (Berkowitz and Philo, 2007), or with multiwavelength UV (Maruno et al., 2021) that have each facilitated precise quantification of AdVs, avoided interference from light scattering and eliminated the dependence on theoretical extinction coefficients for calculating empty/full ratios respectively. Analytical band centrifugation (ABC) (Khasa et al., 2021) that involves sample layer application atop a bulk solvent separates via a self-gradient and thus, as a technique displays higher resilience to systemic effects like temperature and cell alignment variations compared to AUC. While showing potential, more exploration of AAV analysis with ABC and a better understanding of ABC as a function of the bulk solvent properties are necessary for widespread adoption of the technique in quantifying the % full of AAV vectors.

This ability of AUC in monitoring multiple quality attributes highlights its use as PAT; however, the current instrumentation (e.g., Optima AUC by Beckman Coulter) only allows for the analysis of up to 14 samples at once, with <0.5 mL sample volume requirements (Edwards et al., 2020). Additionally, AUC data analysis requires the identification of sedimentation velocity profiles through development work, which can be serotype/particle class specific (Werle et al., 2021), and analysis using software and models such as SEDFIT – that utilizes a sedimentation coefficient distribution model (Burnham et al., 2015; Maruno et al., 2021; Wu et al., 2022a) and others (Edwards et al., 2020). This highlights a need for expertise, which in combination with limited scalability, throughput, and requirement of a high concentration of particles for efficient characterization (Cole et al., 2008; Harding, 2005; Liu et al., 2006) presents limitations that still need resolution for widespread and easy adaptation.

2.3. Electron microscopy (EM)

Scanning- (SEM), Transmission- (TEM), cryo-electron microscopy (cryo-EM), and atomic force microscopy (AFM) have conventionally been used to characterize materials and advances in detection such as enhanced imaging quality via negative staining using heavy metal solutions (De Carlo and Harris, 2011; Mateos et al., 2018) and data analysis have further facilitated characterization of low electron-scattering (biological) particles like AAV vectors and LV (Chen, 2007a; Do Minh et al., 2021; Dukes et al., 2021; Earley et al., 2017; Hendricks and Gao, 2010; Kumru et al., 2018; Li et al., 2020a; McPhee et al., 2006). The presence of the transgene in the full and partial capsids of AAV vectors interacts with the electron signal resulting in varying contrasts between the empty and the partial/full capsids, which can further be coupled with image analysis software to obtain these respective quantities (Dobnik et al., 2019; Fu et al., 2019). However, TEM cannot differentiate particles with fragmented genomes (Burnham et al., 2015). Insights into fundamental particle physiology and kinetics such as temperature-dependent morphology changes (Rayaprolu et al., 2013) or AAV receptor binding interactions have also been facilitated by TEM (Xie et al., 2013) and AFM (Negishi et al., 2004). Cryo-EM has been reportedly used for measuring the size of AAV-DJ particles (Yang and Tang, 2000) and could differentiate empty from full particles (Subramanian et al., 2019). These methods, however, ran on the order of several (7+) hours and needed analysis of large quantities of samples for accurate reporting. AFM has been commonly used for morphological characterization of AAV serotypes and thermal stability (Bernaud et al., 2018) but not directly for establishing CQAs (Bernaud et al., 2018; Chen, 2007b; Zoratto et al., 2021a).

Apart from gaining critical knowledge of genome packaging in terms of full or empty capsids, fundamental insights may not be considered as CQAs which justifies their predominant offline mode of analysis and

reduced applicability as a routine analytical tool. EM instrumentation also has a large footprint with higher sample concentrations (10^7 particles/mL) and longer analysis time by an expert – however, various innovations have been reported for the reduction of (i) time sample preparation, measurement, and data processing, and (ii) the sample volume. Noteworthy examples include a concentration method post negative staining for enveloped pseudo viruses (Zaitsev et al., 2019), robust AAV analysis using an automated, capillary-driven microfluidic device for sample preparation (Hauser et al., 2020). Simultaneously, images and metadata generated by EM-based analytics can definitely serve as mines of information for deep learning and as training sets for other learning models.

2.4. Differential scanning: calorimetry (DSC) and fluorimetry (DSF)

The thermodynamics of stability of biochemical entities can be estimated through their interactions, that are often facilitated by techniques such as differential scanning calorimetry or fluorimetry (Yang et al., 2020), which let us observe changes like protein unfolding, change in enthalpy or entropy with changing temperature. Calorimetry is surpassed by fluorimetry as an effective detection module due to its lower sample volume requirement for similar data resolution and ability for easier sample preparation using commonly available instruments like qPCR.

The value of DSF is in its use for establishing identity of purified vectors especially when accurate antibody-based methods are unavailable for all serotypes (Tseng et al., 2016) and can orthogonally be confirmed by methods like variable temperature charge detection mass spectroscopy (Kostelic et al., 2022). DSF is also sensitive to single point mutations in AAV capsid structure and overall tryptophan content, providing an opportunity to distinguish between serotypes just based on such differences in capsid primary structure (Rieser et al., 2020) and can be extended to evaluate batch-to-batch product consistency (Pacouret et al., 2017). Several instruments (e.g., MicroCal, Malvern Panalytical) are available for DSC measurements in 96-well formats (Markova et al., 2022). Instruments like Prometheus Panta® by NanoTemper or Unchained Labs (Grossen et al., 2023) enable multiplexed assays including DSF, making their at-line implementation possible, however, particle stability and response to temperature may not be necessarily critical to assessing manufactured product quality. This is supported by surveying the applications of differential scanning techniques which highlight its numerous uses in discerning formulation stability in samples at concentrations of $\sim 1\text{E}13$ to $1\text{E}14$ particles/mL, or assessing protein-ligand interactions (Bennett et al., 2017; Lang and Cole, 2017). At the drug substance stage, these techniques have facilitated observing AAV stability at different buffer compositions (Pacouret et al., 2017) and also serotype identity, due to their differences in melting points. (Xiang and Hao, 2023) – which is still a necessary battery analytical test (Bennett et al., 2017; Cole et al., 2021a; Rayaprolu et al., 2013).

3. Emerging techniques with process integration potential

3.1. Particle tracking, flow virometry and other imaging techniques

Understanding particle concentration, biochemical quality, distribution and size can directly inform sizing of purification unit operations and/or therapeutic dosage, and are hence critical to VV manufacturing (Gast et al., 2019a). Particle counting has traditionally been performed using Flow cytometry – that detects particles as events of specific fluorescent labeling, or using light scattering techniques – where intensity fluctuations or angular dependence of particle size help derive particle size through the Stokes-Einstein relationship – overall, often requiring expertise in their respective operation theories (Chen et al., 2022) for data analysis.

Nanoparticle tracking analysis (NTA) is an alternate technique for tracking particles like viruses within the size range of 25–100 nm, with

increasing confidence with larger sizes (e.g., NanoSight, Malvern Panalytical, ZetaView PMX etc.) (Biovan, 2021; Ryu, 2017). This analysis combines particle scattering via a laser beam with specific geometry and imaging their motion and point scatter using diffusion coefficients to estimate their diameter and count, automatically (Heider and Metzner, 2014). NTA instrumentation usually consists of a camera to record particle motion, estimate diffusivities, and thus report statistical size and concentrations. Several reviews have described the use of this technique in biopharmaceutical applications (Lothert et al., 2022; Oshinbolu et al., 2018; Roesch et al., 2021). NTA has been successfully used for LV concentration estimation from purified samples within the range of $5\text{E}9$ to $2\text{E}10$ particles/mL and cross-validated with qPCR and TEM based size analyses with reasonably high correlation ($R^2 = 0.8$) (Kumru et al., 2018). Temperature and buffer viscosity that otherwise impact particle diffusion could be effectively normalized through mathematical manipulations, usually offered by instrument software. Flexibility in employing various detection mechanisms such as dual mode light scattering or fluorescence excitation has simultaneously enabled the sensitive detection and analysis of various particle types, including VLPs, their distributions (Pereira Aguilar et al., 2019) and AdVs at as low concentrations as $1\text{E}5$ particles (Gast et al., 2019a). As typical AAV size approaches the lower limit of detection using NTA, technical advancements in sample preparation such as coupling them with gold nanoparticles have also been looked into, for improved sensitivity and lower scatter (Gimpel et al., 2021). Current instrumentation allows for sample analysis within seconds or minutes, making NTA suitable for at-line analyses in industrial processes. However, these analyses may only suit relatively pure samples from the late purification stages. Conversely, complex samples from early stages may be unfit for NTA due to the noise level caused by smaller scattering particles, which compromise sensitivity (Gross et al., 2016). Recent advancements in microfluidic methods of sample preparation prior to NTA, such as those reported for extracellular vesicle isolation from complex sources (Leong et al., 2022; Tayebi et al., 2020), may expand the applicability of this method in VV biomanufacturing.

Tunable Resistive Pulse Sensing (TRPS, qNano or Exod™ by Izon Bio) accommodates a wider range of detectable sizes, possesses good resolving power for sub-populations but a higher limit of detection in size (~ 40 nm) but has been mostly reported as used for unwanted, large particle quantifications in mAb formulations (Stelzl et al., 2022) and more recently demonstrated for AdV and LV analyses (Heider et al., 2017; Yang and Yamamoto, 2016). Sample preparation comments similar to that of NTA apply to TRPS, tending towards requiring clean and highly pure samples for analysis as of today, thus suitable at late downstream processes. Full-field interferometry-based imaging has also been developed to facilitate label-free counting and concentration predictions of LV, baculovirus and AdV with good correlations with ELISA (Enzyme-linked Immunosorbent Assay) and PCR-based orthogonal methods (Turkki et al., 2021). High resolution imaging (HRI) has been reported for use with various viruses and utilized in making informed decisions within manufacturing processes such as optimizing culture time of infection (Yi et al., 2022), even with adherent cell cultures/microcarriers. While sample preparation for HRI is moderately time consuming, these are opportunities for developing conduits that may include multiple channels for reagent flow/delivery and provide adequate residence times for reaction, enabling at-line application. Single particle measurements (SPM) have also been noted for use with viruses in understanding their physiology (Arista-Romero et al., 2021; Ruthardt et al., 2011). Single AAV2 virus tracking labeled with quantum dots enabled tracking their presence within live cells and has potential to enable virus titer estimations within bioreactors itself (Liu et al., 2020b).

Flow Virometry (FV) emerges from the concept of flow cytometry (FC) while offering rapid detection of virions through non-virus specific fluorescent staining of proteins/nucleic acids on nanoparticles between 10 and 1000 nm and their detection in flow without sorting (Rossi et al., 2015). FV overcomes size limitations of conventional FC through

instrumentation with higher sensitivity, coupled with virus capture using antibody-coated magnetic nanoparticles or scaffolds that additionally also enable differentiating between fully formed virus particles and physiologically similar extracellular vesicles (Bonar and Tilton, 2017; Renner et al., 2020). This method has advantages over methods like TEM with high limits of detection for particle counting as seen with its application with HIV quantification using just ~600 particles/mL (Bonar and Tilton, 2017) and correlates well with qPCR-based genome estimations, even for those with highly irregular shapes like Filovirus between 1E6 and 1E8 vp/mL or Ebola virus (Ricci et al., 2021). FV applied to murine leukemia viruses helped infer that <25% of the signal from its specific capsid protein-based ELISA correlated with its intact virus particle – supporting the anecdotal view in present day lentivirus quantifications (Renner et al., 2020). Guidance on instrument selection, method development, automated statistical data analysis (Cheung et al., 2022) across many applications of this tool have been reviewed. Viro-metric analyses for potency assessment and differentiating morphological changes within test samples indicating product quality issues – as demonstrated with live vaccine drugs, especially using just microliter quantities of samples within 10 min to results highlight the advantage of FV as a PAT tool even on manufacturing floors (Ricci et al., 2021).

3.2. Mass spectrometry (MS)

Due to the importance of mass spectrometry (MS) as a characterization tool that is used across scientific domains, customization of their modules i.e., ionization, acceleration, separation, detection is performed based on the properties of the analyte in question (Büyükköroğlu et al., 2018). Large particles such as VV whose typical molecular weights are upwards of 1 MDa often pose a challenge to native MS techniques, because of their size, biophysical lability and structural complexity for multiple-charge deconvolution (Zoratto et al., 2021b) – needing catered optimization by MS operation. In case of VV, the most reported versions of MS include MALDI-TOF, (nano)-ESI-MS, LC-MS(/MS), Native and Charged CD-MS, nES QR-TOF, and in specific cases – PTR-MS, also in hyphenated forms with other separation techniques like Capillary electrophoresis etc. (Kaur et al., 2022; Serrano et al., 2023). Furthermore, different modes of MS analyses can be qualified as online/at-line or offline analytics depending on the need for sample preparations (Gargalo et al., 2020). It is important to incorporate diligent contextualization and effective management of analytical metadata generated with MS methods as it can provide process insights and an opportunity to serve as training data for deeper learning using AI/ML tools (Lazari et al., 2023; Zhang and Liu, 2018).

Within upstream manufacturing, soft ionizing MALDI-TOF-MS has been successfully implemented for monitoring virus-like particle generation and dynamics from intact Sf9 insect cells (Franco et al., 2010), with the potential for serving the same with Baculovirus cultures. Electrospray ionization (ESI) based-MS was also integrated with a dynamic sampling platform to study bioreactor microenvironments through just ~1 µL samples withdrawal highlighting analytical abilities at-line and robust analyses even with low sample requirements (Chilmonczyk et al., 2021). Beyond the production phase, MS methods for VV molecular weight estimation have been developed, hence translating into capsid identity determination and/or full-empty status.

Combination of two nano electrospray technologies that deploy gas phase electrophoretic mobility (nES GEMMA) and quadrupole reflectron TOF (nESI QRTOF-MS) respectively have been utilized for sensitive AAV8 molecular weight estimations like 3.7 MDa to 5 MDa, and thus insights into empty/full particles respectively (Zoratto et al., 2021b) along with native MS (Strasser et al., 2021b) that has been shown to correlate well with AEX-HPLC readings for the same. Alternatively, Charge Detection MS (CDMS) has been used in conjunction with gel electrophoresis for understanding the mass distribution of genomic content with AAV (Barnes et al., 2021; Pierson et al., 2016). These studies mark important findings of packaged DNA heterogeneity in VVs,

whose biochemistries have been reviewed and concerns on their potential immunogenic effects or reduced product potency have been flagged (Brimble et al., 2023), pushing for their identification as critical attributes to measure in VV production processes. These *partial* species have also been found to be PCR positive yet negatively skew cell-based potency assays (Troxell et al., 2023). Interestingly, both CDMS and nES GEMMA estimated empty AAV8 to weigh 3.7 MDa, while particles with encapsidated genomes were found to have a distribution with a maximum value based on GOI size. Ion-trap MS with laser-induced acoustic desorption and direct charge detection (Ion Trap LIAD-native CD-MS) has been developed for particles between 1 MDa – 1GDa (Lin et al., 2012). Using this, HIV/LV particle molecular weights with two copies of a 9.2 kb ssRNA was estimated to be 3.51E8 Da and was found to be within a 10% mass distribution of theoretical estimates of HIV particles, i.e., 2.67–6.43E8 Da. Both these techniques offer high value for adaptation into total quality control testing batteries for VV drug products to ensure maximum dosage potency for in vivo applications, especially when many particles are needed to produce desired therapeutic effect.

High resolution MS (HR-MS) (Murray et al., 2006), Reverse Phase-TOF-MS (intact protein and digested) (Van Vliet et al., 2009; Zhang et al., 2021), LC-MS (Guapo et al., 2022; Jin et al., 2017), MALDI-TOF (Mary et al., 2019) and HILIC-MS (Liu et al., 2020a) have all been used for capsid protein identity analyses of AAVs and Adv (Orbitrap-MS, LC-MS Peptide Mapping) (Hickey et al., 2023; Snijder et al., 2014). Understanding the landscape of VV post translational modifications is a growing need due to their anticipated effects on post-transduction activity (Gimpel et al., 2021). With samples as low as 0.5 micrograms, PTMs like N-term acetylation and phosphorylation could be estimated using HILIC-MS coupled with fluorescence detection (Liu et al., 2020a), making it suitable for early-stage analyses within manufacturing. For non-intact MS analyses of AAVs, benefits of using pepsin (instead of trypsin) as a digestive enzyme has been found to improve sequence coverage for identifying capsid proteins by up to 70%, while this sample preparation takes <30-min, thus making it suitable for at-line analyses (Guapo et al., 2022). LC-MS/MS methods can also facilitate serotype identity [156] and profiling host-related proteinaceous impurities across various cell lines that can be used to produce viruses or VLPs, of which there have been increasing reports (Lavoie et al., 2021; Sripada et al., 2023; Sripada et al., 2022; Strasser et al., 2021a). MS based methods for the identification of other product-related impurities such as capsid fragments, AAV-interacting complex proteins, nucleolin, nucleophosmin, ribonucleoproteins (Rumachik et al., 2020; Strobel et al., 2015), extracellular vesicles in LV preparations (Do Minh et al., 2021), or rather toxic residual impurities like iodixanol (Pu et al., 2022) used for full product enrichment have also been developed and can be noted for non-critical, offline testing within VV manufacturing space.

A plethora of opportunities within the subject of MS for viral vector growth and production analyses are awaited, and these can be categorized into hardware, software, or data-related advancements. PAT-readiness calls for faster readout, higher throughput and best use of data, the latter of which is critical especially considering the volumes of data typically generated through MS analyses of most types. Movement in this direction is slow yet has begun – as seen with CDMS where chemometric methods and data deconvolution algorithms have been suggested (Kostelic et al., 2021; Marhuenda-Egea et al., 2013), and calls for diligence and focus amongst the scientific community. Research in microfluidic sample enrichment platforms has enabled the reduction of sample requirements thus increasing analytical applicability across purification platform. These work via micro-dispensing of nanoliters range samples onto custom Si vials, hence decreasing the limit of detection to attomole levels and thus far have been for MALDI TOF-MS systems (Ekström et al., 2001). Several other examples of successful MS adaptation as inline/online tools exist today, for instance, the design of a microdevice for real-time sample treatment enabled coupling bioreactors with an ESI-MS system to gather insights into the bioreactor

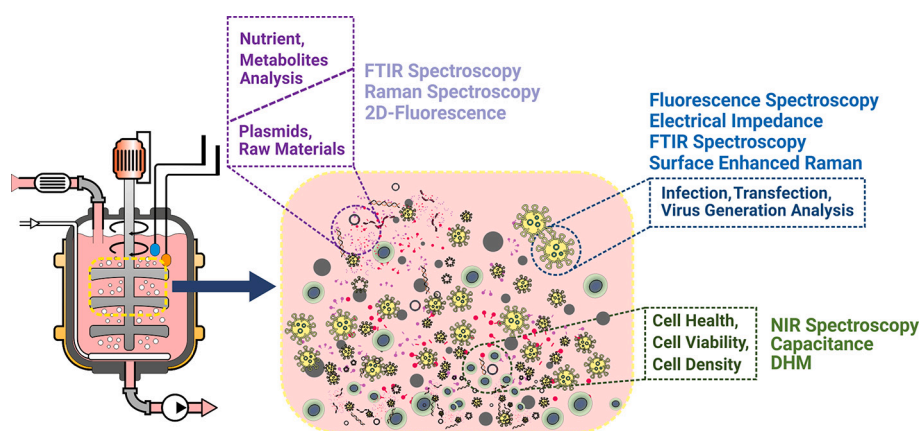


Fig. 4. Spatial description of sample classes (media – nutrient, cells – chassis, viral vectors – product) typically met in upstream viral vector biomanufacturing, and real-time enabling analytical techniques being developed as PAT tools for each category. The temporal nature of cell growth and virus production occurring within the same unit operation ideally calls for an analytical technique that can monitor multiple attributes. Advances in using Fluorescence spectroscopy and Electrical impedance-based methods for both cell health monitoring and viral production-related analyses have been specifically developed for AAV and Lentivirus analyses. Further development opportunities from current limitations include techniques for sensitive and robust signal transduction, suitability, and robustness across various viral vector products – accommodating for differences in production strategies (lytic versus enveloped/secreted viruses), wanting methodologies for effective spectral data analysis and space-effective, sterile hardware.

microenvironment (Chilmonczyk et al., 2021), and HPLC methods coupled with MS as the last dimension of analysis (Piendl et al., 2019). MS has been reportedly also been ‘hyphenated’ with several non-traditional methods like capillary electrophoresis (Zhang et al., 2018) or Chip-HPLC (Piendl et al., 2019), to preface analysis with separation based on charge or other molecular properties. Several CE-MS methods have been reported in scope of VVs (Chen et al., 2021; Stolz et al., 2019) for capsid identification and PTM mapping. A review on challenges and opportunities in MS development for AAV analytics specifically has also recently been published (Serrano et al., 2023).

3.3. Mass photometry (MP)

Mass Photometry (MP) or interferometric scattering microscopy quantifies the scattering of light from individual biomolecules in solution to find their molecular mass, with limited sample volumes and no staining requirements. Contrary to conventional light scattering – especially techniques like DLS (discussed in Section 4.4), MP works by reducing reflected light from an illumination source while simultaneously capturing the scattered light from the particles of interest with minimal disturbance (Cole et al., 2017). MP has an expansive scope, enabling the study of various biomolecule categories like proteins (Higuchi et al., 2021), DNA (Li et al., 2020a), RNA, large macromolecular complexes (Torres-Ocampo et al., 2020), and even small viruses such as AAVs (Refeyn, 2022). Recently, MP has been used to analyze the content ratio of rAAV5, using 1 mL sample volume with AAV concentrations in the range of $1\text{E}12$ to $1\text{E}13$ cp/mL but provided readouts in only a few minutes (Refeyn, 2022), thus encouraging its use close to process lines. Photometric methods have been pursued to address both technical challenges, i.e., reduction in signal noise and logistics – in the form of sample volume, while achieving similar resolution. In addition to quantifying the different empty, partial, and full fractions, Wu et al. (Wu et al., 2022a) used MP to quantify the size of the encapsidated genome using the contrast values of the images collected during production. Wagner and colleagues successfully computed the molar mass and assessed AAV subpopulations in samples having capsid titers as low as $8\text{E}10$ cp mL^{-1} , achieving a CV of <5%, all utilizing a mere 10 μL total sample volume.

MP offers dependable molecular mass measurements within a range of 30 kDa to 5 MDa, with the AAVs falling within the 3–5 MDa, and hence is a wide range suitability limitation. Additionally, the technique is highly sensitive to impurities, restricting its usage to pure or freshly

prepared rAAV samples. Its capability to measure multiple samples within a brief timeframe provides MP with an edge over the relatively more time-consuming techniques such as AUC, CDMS, and TEM (Wagner et al., 2023).

3.4. Capillary electrophoresis

Capillary electrophoretic methods (CE) harnessing molecular properties as a function of electrical field strength have been widely utilized for biological product analyses and thus in biopharmaceutical characterization (Shah et al., 2022). CE is usually applied with SDS based migration, separation of charged species using ampholytes or in denaturing conditions (Kaur et al., 2021; Kumar et al., 2022; Placidi et al., 2020; Shah et al., 2022). Kumar et al., (Kumar et al., 2022) provide a comprehensive list of protein biologics and critical attributes in their manufacturing measured using these methods, while Fernandes et al., (Fernandes et al., 2022) have provided an overview of how they are implemented across the viral vectors space across particles with and without envelopes.

AAV analysis using CE has been used for profiling genomic content and assessing sample purity and potency. CE coupled with laser induced fluorescence-based detection (LIF, available as Sciex Biophase 8800 or PA800, Maurice™ by BioTechnie, Agilent Zip-Chip etc.) has been observed to provide required sensitivity by increasing particle detectability for analysis (Lock et al., 2020). CE-SDS is a standard method for analyzing the ratios of viral proteins and thus identity, even across various AAV serotypes. With method optimization (Li et al., 2020b), this has been observed to estimate AAV purity at $\sim 1\text{E}13$ vg/mL concentration, within 30 min (Aebischer et al., 2023). However, it is still generic to (sero) type of vector and unsuitable for profiling product related impurities like clipped or modified variants. Furthermore, with specific sample digestion, LIF detection also enabled AAV payload size and integrity analysis (Luo and Guttman, 2020), which has been successfully demonstrated for genomes in the range of 2.3–4.7 kb. The sensitivity of LIF for viruses was also harnessed in differentiating viral RNA payload from residual host DNA to quantify the latter as a critical process quality attribute with high confidence up to 1 ng/mL (Wang et al., 2023). While virus particle heterogeneity can be structural differences due to protein sequences or PTMs, CE as isoelectric focusing (IEF) can resolve capsid compositions due to their resultant pI differences (Ramírez et al., 2023). With current instrumentation, IEF can focus AAV samples within 7 min (X. Z. He et al., 2023). Additionally, the optimized use of ampholytes in

IEF sample preparation have also been observed to enable detection of filled capsids within a sample as low as 65 nanoliters (Li et al., 2020a; Wu and Heger, 2022), hence making IEF suitable for analyses at manufacturing floors.

AdVs have been characterized using both capillary and zone capillary electrophoresis (van Tricht et al., 2019; van Tricht et al., 2017), within vaccine manufacturing spaces with appropriate orthogonal validation. Lentivirus titer determination using p24 signatures and purity at concentrations of 1E9 TU/mL and above have been successfully determined using CE-SDS and described as a potential tool for assessing batch variations of product (Fernandes et al., 2022; Li and Guttman, 2021). An interesting application of chip-based CE has been shown in sample fractionation followed by establishing payload identity using molecular beacons – specific hairpin shaped complementary DNA strains detected using nano electrospray based molecular analysis (Pitchler, 2020). As mentioned in Section 3.2, CE is also used as the first separation dimension for MS and/or LC methods (Aebischer et al., 2023; Chen et al., 2021; He et al., 2023a; Metzger et al., 2009; Stolz et al., 2019; Wu and Tang, 2020), also while being implemented at micro-fluidic scales. However, current instrumentation for this assay has been associated with low sensitivity due to narrow capillary width that may not pave way for enough light for detection (Pinto et al., 2021). Since then, further advancements have been made in the development of microchips with varying capillary widths on a wafer (Ou et al., 2020), pending evaluation with viruses and multi-capillary electrophoresis platforms (Darling, 2022) that are also suitable for high concentration samples like drug product analysis with the advantage of implementing at processes directly due to analysis scale.

4. Emerging inline, online and at-line techniques

Rathore et al., recently published their prudent ideas on the composition of next generation analytical platforms as those incorporating advanced methods, automation coupled with efficient data analysis, ultimately enabling real-time monitoring while being resource-lean (Rathore and Sarin, 2023). This section synonymously focuses on recent spectroscopic and light scattering tools that address VV product-specific monitoring needs – within upstream manufacturing (Fig. 4), followed by an expanding landscape of biosensors. Monitoring consistency in viral assembly and therapeutic gene packaging at all stages of manufacturing is key in understanding safety and efficacy of the biologic within the manufacturing process and may also aid in decision making (Green and Lee, 2021). Variables specific to upstream manufacturing that require special considerations with respect to developing analytics are suitability with both adherent and suspension systems, and hydrodynamic similitude between sensors that enable monitoring these processes across scales – for consistency (Panchalingam et al., 2021). Additionally, cellular environments challenge tracking unique virus-specific biomolecular signatures (Warnock et al., 2006). Given the diversity of VV expression systems and production methods (Merten et al., 2014; Warnock et al., 2006), authors have hence suggested viewing analytical challenges as process-specific and product-specific, rather than source-specific (van der Loo and Wright, 2016).

4.1. Fluorescence-, dielectric-, infrared-, Raman spectroscopy

Spectroscopic techniques have successfully been implemented for bioreactor monitoring through insertable probes designed with transmission and detection elements in suitable formats, and as single-use or autoclave-resistant (Henry Weichert et al., 2014). Many emerging techniques are preceded by the success of Raman Spectroscopy, now ubiquitously used in the pharmaceutical setting for real-time release testing, offering a lot to learn from (Esmonde-White et al., 2017). Fluorescence spectroscopy has been used to monitor the growth and AAV production of baculovirus-transfected Sf9 insect cells (Pais et al., 2019) and is a solid example that creatively ameliorates interference

from background noises generated in bioreactor environments by media components, or due to cell density and inherent autofluorescence. Generally, spectroscopic tools often open doors for large amounts of data, whose efficient analysis through multivariate tools, chemometrics, artificial intelligence (AI) and/or machine learning (ML) can lead to gaining increased process insights (Lussier et al., 2020), also discussed in Section 5. In this case, a unique method using two-dimensional spectra to correlate with process parameters has been reported using synchronous spectra, i.e., a constant differences in wavelengths – in this case 20, 80 and 140 nm between the range of 250–520 nm, similar to an idea previously reported used in protein concentration monitoring (Ghodbane et al., 2019; Sripada et al., 2020), for developing an online method. ML algorithms (Artificial Neural Networks (ANN), Partial Least Squares (PLS)) were used to further optimize fluorescence spectra preprocessing to observe precise predictions of cell concentration, viability and AAV titer. Such information ultimately helps correlate cell health to viral titer or potency and make decisions on the best harvest time etc. Intrinsic fluorescence which is common with many culture media and metabolites has been shown overcome through the use of Raman (e.g., Kaiser Rxn3) – that bases tracking on molecular signatures, or special sensors (Chen et al., 2020) and high power techniques like 2D fluorescence (Pais et al., 2019). This also highlights an opportunity for fundamental studies correlating growth/production changes to specific cellular pathways such as glycolysis, gluconeogenesis or metabolic cycles such as TCA, fatty acid oxidation, oxidative phosphorylation – all of which are also associated with virus production kinetics (Rasmussen et al., 2011; Sánchez-García et al., 2021). Fluorescence was utilized for predicting HIV VLP concentration in HEK293 cultures (Helgers et al., 2022), suggesting transferability to lentiviral systems (Do Minh et al., 2021).

Dielectric spectroscopy for mapping live to dead cells in a culture (Prodan et al., 2008), cell integrity (Justice et al., 2011), and even growth rates or product expression (Reyes et al., 2022) and capacitance for cell culture monitoring are available and used as PAT, bolstered with real-time analytics for bioreactor environments. These are available as probes (e.g., Aber, Hamilton Incyte) and have been applied with both AAV and LV productions in HEK293, operated as both suspension and adherent cultures (Yardley et al., 2000). Kamen et al., noted permittivity differences using a specific parameter $d(\epsilon)_{\max}$ due to cell physiological changes upon LV infection (Ansorge et al., 2011), which was measured robustly across different process phases including perfusion. The same metric additionally facilitated tracking lipid-enclosed volume fractions of Sf9 (insect) cells, suiting correlation with production of enveloped particles, thus virus expression (Zeiser et al., 1999). While possible with adherent and microcarrier cultures (Petiot et al., 2012), linear correlations have only been found possible at fully colonized conditions, but perhaps not still reliable due to changing cellular morphology (Cole et al., 2015). Within AAV production, permittivity measurements that typically exist between frequencies of 500 kHz – 2 MHz were found to be lower (300–442 kHz) in non-infected Sf9 cells (Negrete et al., 2007), highlighting the ability of this technique to directly pick on sensitive physiological changes such as differences in membrane polarization states. An increased training set of this data with orthogonally measured “disturbances” in content ratios of particles enabled the development of a robust multiple linear regression model for AAV concentration prediction. True process optimization can be carried out with such information upon identifying the intersection of best cell health, viability, and process economics (Pais et al., 2020a).

Fourier transformed IR while developed and used for nutrient and metabolite analyses, and implemented with adherent cells like Vero (Petiot et al., 2010), has also been used to identify infectious viruses like HIV etc. (Santos et al., 2020), making it amenable for LVV monitoring using similar hardware and software (e.g. ReactIR series by Mettler Toledo). Near IR for optical density measurements has also enabled measurement of cell culture turbidity generally correlated cell densities. With competent data analysis, transmission and backscatter of light measured via probes (e.g., Hamilton Dencytee, Mettler InPro, Sartorius

BioPAT etc.), has been used to map living and dead cell populations distinctly *in situ*, as used in insect cell and AdV producing Vero cell bioreactors (Poetner, 2020). NIR however is less suitable in micro-carrier/adherent cell growth environments (Poetner, 2020). While enabling non-invasive methods for multi-analyte analyses, IR monitoring method development must be carried out carefully to overcome limitations such as water interference, multi-analyte monitoring especially in cultures with high cell densities and robust real-time data collection even at high agitation/aeration conditions (Lourenço et al., 2012). Methods ensuring sensitive detection of differences between viral and cellular components need to be developed.

Helpful guidelines for correlating membrane capacitance and intracellular conductivity across LV, baculovirus and influenza virus production cultures are also available in literature (Petiot et al., 2017). Capacitance measurements for biomass monitoring in cases of virus vaccine candidate manufacturing has also harnessed the polarizability of enveloped viruses and shown their distinct monitoring at a capacitance of 15 MHz, correlation with viral titer measured offline using TCID₅₀, while host cells were trend at 580 kHz across scales (Yi et al., 2022). Multivariate methods like PCA using multiple measurement frequencies have also been utilized to create universal capacitance models (Konakovsky et al., 2015). Current drawbacks and future opportunities of this technique for monitoring cultures stem from poor measurement at low-viability conditions, which may be overcome with better data analytics or even hardware development. Surface Enhanced Raman Spectroscopy (SERS) based lentivirus titer determination has been developed (Morder et al., 2022), with the potential for design as a real-time probe. ViroFluidics – that combines principles of FV and microfluidics in LV cultures provides a case study for its inline readiness by sampling particles at relatively low concentrations (~1E4/mL) while abetting monitoring of the culture from a single cell level.

Developing spectroscopic tools for bioreactor monitoring requires the coming together of molecular biologists, process engineers, analytical scientists and chemometricians – which is exciting but also shows potential challenges associated with raw data discernment or troubleshooting. Furthermore, any sensors online that are integrated into process streams may also require extensive testing to prove stability even with solvents and high temperatures that may be used during cleaning, currently a limitation with NIR (Vann and Sheppard, 2017). However, it is important to remember that deploying existing methods within routine VV manufacturing right away can help generate meaningful training data sets for continued learning and enabling soft sensing – that can use ML models for predicting key process attributes. Their developability depends on finding robust property predictor variables using the sensing elements, forecasting multiple attributes variables using these variables which tie into diligent data logging and contextualization. Current knowledge of mechanistic models for VV production – such as AAV have been discussed elaborately (Iglesias et al., 2023).

4.2. Content imaging (microscopy-based)

Differential digital holographic microscopy (DHM) has been used for soft sensing and CQA measurement within upstream processes, using over thirty optical attributes from a bioprocess. It is an online, label-free, quantitative phase imaging technique, that constructs holographic images based on phase differences of cells under scrutiny. DHM coupled with forward stepwise regression analysis has enabled viable cell concentration monitoring and intracellular- and extracellular AAV titer predictions in Sf9 cultures (Pais et al., 2020b). More recently, Altenburg et al., built on this work by demonstrating the same technique in baculovirus infections, where they employed machine learning models to non-invasively gather information on infection events and metabolic state of the cells (Altenburg et al., 2023). The method showed robustness across differences in culture conditions such as different MOIs or chemical additives. Other imaging techniques like 2D fluorescence lifetime imaging has been developed cell health and metabolites

assessment at-line in other biopharmaceutical cell lines, (Sternisha et al., 2021), and flow-cytometry approaches have also been shown for use in real-time cellular analyses at single cell levels (Broger et al., 2011; Kuystermans et al., 2016). Sampling conduits to enable online analyses by recirculating a bypass stream of bioreactor contents has also been discussed in the context of enabling imaging based PAT (Gargalo et al., 2020).

Recently, brightfield microscopy has also been developed with the use of convolutional neural network machine learning algorithms as a relatively rapid, highly precise infectivity assay for viruses, thus improving result turnaround (Dodkins et al., 2023), compared to standard assays (TCID₅₀, Plaque Assay etc.) that often take days. This method has been tested with AdV and trained with other DNA-based, RNA-based and enveloped viruses. Automated, periodic (~2 h) temporal imaging of LVs with immunological stains could determine infectious titer when incubated with HEK293 cells in real-time using the Incucyte® platform (Labisch et al., 2021). This upgrade serves as a good example for innovation within a currently manual, highly error-prone state of assays – that also use data analysis templates, facilitating at-line analysis. As a thought for the future, criteria for technology development should be coupled with a thorough analysis of analytical design space to understand gaps in achieving both improved process knowledge (Reyes et al., 2022) and enable process control.

4.3. Analytical chromatography

Implementation of HPLCs as online methods has been around since developments in mAb manufacturing (Tiware et al., 2018). A successful example of bioreactor monitoring using analytical HPLC (aHPLC) for measuring various CQAs is from Bristol Myers Squibb, that spearheaded many efforts to translate lab-scale PAT to clinical/manufacturing floors. They demonstrated sterile, automated, scheduled sampling (using the Fisp® probe, Flownamics) delivered to a UPLC titer measurements and also developed a (Infinity) 2D-LC method for multiple measurements including product titer, variants (IEX, SEC) and amino acid compositions. (Chemmalil et al., 2021; Chemmalil et al., 2020). Their experience can teach us the multitude of aspects that require consideration and optimization to facilitate HPLCs as online PAT, such as sterility-ensuring interfaces, sampling systems, software integration/communication between electronics for real-time injections, and their overall connectivity with local or distributed control systems (for process control, timely data logging, assimilation, and inter-operation communication). These considerations are also discussed in Section 6. Specific LC analyses in high-pressure or ultra-pressure modes have helped track/map signature molecular properties of analytes like charge, size, hydrophobic/philic interactions etc., for effective separation of various analytes across domains. Today, they are also integral within analytical testing batteries for enabling process-related and product related impurity measurements with low sample volumes (Rappold, 2021). Within VV characterization, aHPLC methods have been developed for understanding capsid property distributions (IEX-HPLC etc.), size base distributions of virus products and proteinaceous impurities within samples (SEC-HPLC, SXC-HPLC etc.), and even residuals testing for safety (RP-HPLC, SEC-HPLC etc.) (Sripada et al., 2024; Transfiguracion et al., 2011). Their method development entails optimizing mobile phase composition, sample preparation techniques, detection methods etc., many of which have been pursued for analyzing viruses.

Several reviews effectively summarizing aHPLC methods used for VV characterizations have been published (Fekete et al., 2023; Gagnon et al., 2021; Kramberger et al., 2015). Gagnon et al. conveyed the importance of “accommodating the reality” of these analyses which requires thorough understanding of their physical, chemical, and biological natures. Aspects such as capsid primary sequences, payload size, envelope presence, impurities based on process stage and many more – significantly help in finding the best analysis conditions (Fekete et al., 2023). For example, the knowledge of tryptophan residue abundance in

AAV capsid proteins (UniProt ID: P03135) makes it possible to harness the particle's intrinsic fluorescence as a detection method without intercalating dyes that also bind to nucleic acids. The Guillaume group reported a variety of aHPLC methods for AAV analyses, including IEX-HPLC for empty/full particles, SEC for intact AAV concentration or particle counting (in tandem with light scattering detectors), HIC and HILIC – also coupled with mass spectrometry for viral protein analysis, including peptide mapping and for post-translational modifications (Aebischer et al., 2023; Fekete et al., 2023). AEX-HPLC methods for estimating LV, AdV (UV-based detection) and Baculovirus (using SYBR green dyes) particle concentrations are available for estimating high concentration ranges of each (Transfiguracion et al., 2020; Transfiguracion et al., 2011; Whitfield et al., 2009), which although need a standard curve – circumvent sample dilution, thus suiting use in downstream. A reverse-phase method for the identification of 'empty' AdVs using the presence of an indicator (capsid protein VIII) has also been reported (Vellekamp et al., 2001).

Understanding the physicochemical nature of VV within manufacturing helps understand possible variations – which is indispensable for the design and execution of advanced analytics that are precise, exact, have considerable detection ranges/limits and can be integrated into processes for real-time measurement. Some opportunities for aHPLC can be materials related (stationary phase and its design) or from automation. For e.g., monoliths have been used for AAV particle concentrations or genome integrity analyses (Christler et al., 2020; Lock et al., 2012; Luo and Guttman, 2020; Perry and Rayat, 2021; Whitfield et al., 2009); and other membrane based columns for enabling steric exclusion chromatography, that uses PEG to separate virus and other proteinaceous components effectively (Labisch et al., 2022; Marichal-Gallardo et al., 2021). Ultra-short columns (10–20 mm) have been designed for large molecule analysis to improve dead- or holdup volume issues while sustaining high pressures (Fekete and Guillaume, 2023) – however of which no reports pertaining to VV are available. Along with stationary phase particle size that directly influences resolution, pore size – especially with viruses that are relatively much larger than proteins (Fekete et al., 2023), and method details such as sample loading, load flowrate (Gagnon et al., 2021) or mobile phase must be carefully optimized for VV analyses, especially due to the polymeric nature of the vector itself and the types of interaction it can promote with the phase (Hannes Ausserwöger et al., 2017). Additionally, substrates that constitute the chromatographic particles can be used to modulate surface interactions or avoid non-specific interactions (Fekete et al., 2023). An interesting question for consideration is also if determining infectivity/ transduction efficiency via aHPLC will ever be a possibility.

4.4. Light scattering (multi-angle and dynamic – MALS and DLS)

MALS, DLS and static light scattering have been extensively used for characterizing biologics including mAbs (Minton, 2016), in conjunction with fractionation techniques based on size, mass or charge (Amartely et al., 2018; Wyatt, 2013; Young, 1981), for particle stability, identity and composition analyses due to its sensitivity to small molecular weight differences. MALS enables the physical analysis of VV heterogeneity from purified VV intermediates (Perry and Rayat, 2021; Wright, 2014) including deriving vector identity by size/weight and quantification of content ratios. LS detectors placed downstream of an HPLC (IEX, SEC etc.) or other separation modules like FFF make automation and online analysis possible. Typical MALS instrumentation involves a separation module and an LS detector coupled with UV and/or refractive index-based detection (Dawn™ series by Wyatt, Lens3™ Tosoh etc.) – which together provide a superior ability to infer multiple quality attributes via empirical relationships. MALS can utilize clarified cell culture fluid as sample (Bousse et al., 2013), however potential interference with the media/buffer needs prior testing.

Particle concentrations of up to $\sim 2.5 \times 10^3$ particles/mL for AAV and up to $\sim 1.5 \times 10^9$ particles/mL for LV have been previously detected with high confidence – post SEC and FFF respectively (McIntosh et al., 2021; Seal et al., 2021; Sripada et al., 2024). SEC-MALS has been reported as a versatile, one-stop tool for AAV analysis for capsid size, content, stability and integrity assessments in addition to aggregation (McIntosh et al., 2021). Importantly, as not all vectors are known to carry a consistent genetic payload (François et al., 2018), packaging efficiencies may be obtained using this technique due to its ability to estimate molecular weight distributions. Multi-angle DLS has also been developed for AAV quality analyses (Cole et al., 2021a). A benchtop Zetasizer tuned at 633 nm could differentiate full and empty AAV particles, even from in-process intermediate samples, with just 20 μ L samples (Arunkumar and Singh, 2021). A recent comprehensive report of method development of IEX coupled with MALS for AAV serotype detection demonstrates sensitivity of the technique while also highlighting the possibility of identifying truncated AAV particles (Wagner et al., 2022). Additional refractive index detection with MALS also enables the analysis of HIV gp140 oligomers and HIV VLPs online (Barackman et al., 2004; Pereira Aguilar et al., 2019). FFF is a preferred separation mode for larger particles (>100 nm) (Giddings et al., 1977). Interestingly, FFF-MALS could resolve a test sample containing AAV and AdV5 impurities (from upstream) with significant retention time difference (Bou-Assaf et al., 2020) and also allows AAV aggregate quantification, a safety CQA. Simple method development and qualification of AdV samples could be performed using polystyrene beads with similar properties (McEvoy et al., 2011). McEvoy et al., utilized LS information from an asymmetric FFF mode of operation to numerically differentiate influenza virus monomers from aggregates, within 30-min in total (Bousse et al., 2013).

DLS has been used for particle counting, distribution and size estimation of vectors across Adenoviridae (Hohl et al., 2017) and Retroviridae (Kumru et al., 2018) families including AAV and LV (Cole et al., 2021b). An additional electrophoretic scattering module confirmed that particle integrity remained unchanged during analysis, thus increasing reliability of DLS systems (Cole et al., 2021a). With adequate detector power, sample volumes lesser than 100 μ L have been used to estimate $>1 \times 10^9$ AdV particles with only a 5% difference from qPCR measurements; however, effects of high laser intensity needs evaluation with enveloped or labile vectors like LV, whose titers may vary greatly (Carbonaro-Sarracino et al., 2020; Soldi et al., 2020). A choice of both high-throughput plate reader formats for at-line or online analysis (Mobius™, WyattQELS™) for DLS is available.

MALS may have low resolving power to differentiate particles with similar physical sizes, especially if prior separation methods are ineffective or within chemically complex samples, challenging its widespread adaptation (Some et al., 2019). Moreover, method development and data discernment with LS also calls for expertise in scatter theory. Capabilities aside, current literature shows confident estimations from only across small ranges of VV product concentrations, while method validation challenges within manufacturing are unseen. However, within the current accelerated GT landscape, MALS supersedes the requirement of intensive, time-consuming biochemical assays like analytical ultracentrifugation (AUC) etc., especially for AAV content ratios, is online/at-line ready in its current state and its dependence on first principles circumvents extensive property correlations.

4.5. Chemical and biological sensors

Though many offline assays within VV manufacturing today are vital, several setbacks such as assay time, manual data collection, large sample volumes, room for human error, and often, complex instrumentation associated with them can make analyses time-intensive (Petiot et al., 2017). Sensors with rapid measurement capabilities can effectively address these challenges and offer improved sensitivity, multiplexing capabilities, and real-time monitoring while operating

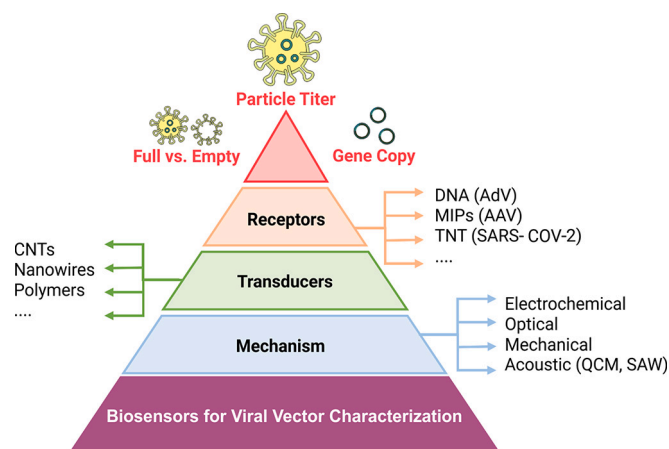


Fig. 5. A synopsis of the different aspects of biosensor construction that require consideration for development within its components for analyzing viruses, improving sample clarification, detection sensitivity and reception - highlighting examples within each category (CNT: Carbon Nanotubes, TNT: Titanium Dioxide Nanotubes, SAW: Surface Acoustic Waves).

with smaller samples and reduced reagent volumes – ultimately to integrate them into VV manufacturing (Mandenius and Gustavsson, 2015). There is possibility of adopting ideas for sensing from existing virus diagnostics, but direct translation may not be feasible due to factors such as differences in sample media based on host cell line or process phase, differences in analysis scales, suitability for integration into process lines and differences in specific analytes. We thus summarize chemical and biological sensor strategies reported for VV analyses in this section (Fig. 5), and characterize them by the type of their detectable signal.

4.5.1. Electrochemical

Electrochemical sensors estimate chemical changes through electrical measurements, like harnessing material dielectric properties – that are obtained by introducing multiple voltage frequencies and measuring the resulting current. Electrochemical Impedance Spectroscopy (EIS) or dielectric spectroscopy has been used for bioreactor monitoring and even virus infection event analyses as discussed in Section 4.1. Many advances with EIS detection also involved substrate definition, such as acrylate and methacrylate based Molecular Imprinted Polymers (MIPs) for (adeno)virus identification (Gast et al., 2019b), AAV epitope screening (Piletska et al., 2020) and carbon nanotubes to interact with capsid protein (Vadlamani et al., 2020), to achieve sensitive analyte detection. A voltammetry-based label-free DNA-based biosensor has been designed to detect infectious enteric AdVs (Song and Jin, 2015), using a polypyrrole film deposited on Si nano-pores and functionalized with a complementary probe DNA to AdV type 40 and 41. Adsorbed DNA was found to correlate linearly with conductivity change, thereby laying groundwork for developing online biosensors for DNA quantification as a substitute for PCR assays. Rapid and ultrasensitive response of EIS-based detection of virus using functionalized Boron-doped diamond electrodes has been reported, as well (Nidzworski et al., 2017).

4.5.2. Optical

Optical sensors are widely used in the form of imaging or localized surface plasmon resonance (SPR) (Damborský et al., 2016), that enable label-free and rapid monitoring of biomolecules and their interactions, in comparison with ELISA or MS based methods, making it an invaluable member of the PAT arsenal (Masson, 2017). An SPR biosensor functionalized with an ion-exchange surface was designed to measure rate of baculovirus sorption, product-derived impurities such as damaged viral particles and general process related impurities (protein, DNA, and endotoxin) (Vicente et al., 2010). Binding and elution at various buffer

compositions were tested with these viruses to obtain adsorption isotherms, enabling a mechanistic understanding of their interactions, thereby aiding early-stage process development. A sensor immobilized with secondary antibodies against AdV on the chip surface and another instance of imprinted AdV sensing (Abad et al., 2002) were also developed for AdV quantification circumventing the need for prior sample labeling (Abadian et al., 2015). While this sensor could operate continuously and be reused up to 20 times, its fabrication required diligence and pretreatment with a known concentration of hexon-specific antibody, for up to ~4 h of construction time per piece. Changes in reflectivity were monitored to estimate AAV5 particle concentrations as an at-line SPR-based biosensor (Cui et al., 2019). SPR biosensors equipped with MIPs (vancomycin-MIP and MS2-MIP) (Altintas et al., 2015) or ion-exchange surfaces (Roper and Nakra, 2006) have also facilitated increased biochemical and mass transport-related understanding of AdVs. Utilizing microfluidic components within biosensor designs has been adopted for virus detection applications, like AdV (Jin et al., 2018). Here, analyte DNA was extracted in situ using a microfluidic system followed by isothermal solid-phase DNA amplification and nucleic acid detection via resonance wavelength shifting, leading to 100-times greater sensitivity than qPCR. Such label-free, real-time monitoring systems effectively overcome disadvantages with PCR like high costs and expertise, making it a promising PAT application.

Fluorescent optical sensors ranging from fiber optics to more complex sensors as PAT tools have been reviewed (Harrison and Chauhan, 2018). The former has limitations in measuring large analytes volumes and may require coupling with fluorescence microscopy for analyses, with the added possibility of damaging biological molecules or cells due to the puncture force of the employed tips. This brought about a transition to fluorophore-utilizing sensors (ratiometric or nano-sensors) that have high sensing capabilities due to their small size and are capable of quantitative measurements due to their inert matrix. Guerreiro et al. developed a genetically encoded switch-on sensor activated by viral proteolytic activity to measure infectious AdV titer (Guerreiro et al., 2019), that reduced infectivity measurement time from up to 14 days with plaque-forming assays to just two. LV particles (HIV-1) were also successfully detected with the same sensor by modifying the cleavable sequence, thus ensuring its wider potential for use within VV biomanufacturing. An optoelectronic sensor fabricated for fowl AdV detection exhibited $>100\times$ sensitivity (LOD 8.75 PFU/mL) than ELISA through the use of FAdV antibodies immobilized onto graphene quantum dots on carbon electrodes, resulting in electrical signal changes measured under UV irradiation (Ahmed et al., 2018). Fluorescence-based sensing, though sensitive, is subject to limitations arising from probe degradation and signal deconvolution, particularly in complex and dynamic samples such as cell cultures and bioreactors. In biological matrices, common limiting factors include autofluorescence due to inherent photophysical properties of biologics, signal-to-noise ratio reduction from scattering media, and potential cross-reactivity with background constituents leading to false positives (Leiner, 1991; Sternisha et al., 2021). A distinctive challenge in real-time inline monitoring using fluorescence-based sensing is the dynamic evolution of the analyte matrix. Changes in pH and dissolved oxygen, inherent to bioreactor operation, significantly impact the quantum yield and spectral response of fluorescent probes, presenting a considerable challenge to overcome. Additionally, fundamental challenges in probe selection encompass non-specific quenching, photobleaching, and degradation over time, influenced by factors such as fluorescence decay during the analysis period or the physical characteristics of the sample, such as pH and concentration. Lastly, while not problematic for at-line or offline assays, the inline and online introduction of fluorescent probes may induce cytotoxic effects or, at the very least, pose an additional burden on downstream purification and product processing. Addressing these challenges requires a nuanced approach to design fluorescent probes, improve selectivity, and enhance the robustness of fluorescence-based sensing in complex bioprocess fluids and environments.

4.5.3. Mechanical

Microelectromechanical systems (MEMS) have been used for many high-resolution chemical sensing applications (He et al., 2023b). There is potential for their use in process analytics due to their exceptionally high resolution, world class limit of detection (zeptogram) and near immediate response time (ms). Successful integration of MEMS into processes also requires addressing critical challenges like ensuring device robustness and reliability in industrial environments, mitigating sensor fouling, drift, and cross-sensitivity issues and availability of advanced data analytics to extract actionable insights from generated high-resolution data (Workman et al., 2011). Besides, MEMS hold promise in advancing PAT due to their high sensitivity, generated from their property of detecting changes in mechano-physical parameters like forces, motion, masses, etc. resulting from biomolecular interactions. MEMS have no labeling requirements and can operate in various phases (Polacheck and Chen, 2016; Rodahl et al., 1995; Tamayo et al., 2013) – also suggesting application advantages with biomanufacturing systems that may involve changing buffer compositions. A suspended nano-mechanical resonator (SNR) has been reported to measure the AAV mass in the zeptogram (10^{-21} g) range (Katsikis et al., 2022), using a hollow cantilever vibrating at its resonant frequency that changed with additional mass. This sensor could also measure relative masses of full (heavy) and empty (light) particles and quantify (i) the % of full capsids, (ii) average DNA content, and (iii) number of aggregates using SNR, working as a true multi-attribute method.

Acoustic transducers or piezoelectric elements can also detect mass changes and generate frequencies through an oscillator circuit. This gives biosensors real-time measurement capabilities as they convert changes in mass due to interactions or membrane properties (visco-) elastic, or dielectric properties) into frequency changes. Quartz crystal microbalance (QCM) sensors are a widely known example of this category, that consist of a thin AT-cut quartz chip with two metal electrodes functionalized with analyte specific receptors, also used for virus detection and quantification. (Afzal et al., 2017). Various surface chemistries have been investigated for building QCM sensors including functionalization with antibodies, MIPs, aptamers, etc., enabling highly selective target capture. For example, a sensor to detect viral RNA encoding G protein was designed using three functional group decorations like thiol, amine, or biotin linkage, to determine biotin as the best (Hong et al., 2010). This sensor could be regenerated 32 times and had detection capabilities as low as 1.6 nM. However, challenges with stability of surface modifications or their limited regeneration remain. A detailed review of QCM biosensors for virus detection has been published (Afzal et al., 2017). Future developments in device design may also focus on effective data collection and transfer, especially those hands-free (Rathore et al., 2022).

5. Multivariate data analysis (MVDA) and machine learning (ML) for viral vector analytics

Availability of the real-time process analytics permits process control and also the execution of semi- to fully continuous manufacturing processes (Lee, 2021), as seen within biopharmaceutical manufacturing over the years. Inline/online PAT tools potentiate not only feedback but also feed-forward control and thus product release based on real-time information (Singh et al., 2015), thus streamlining product output and help avoiding batch failures (Gerzon et al., 2022). Specifically, in VV manufacturing, process control though currently contingent on the development of robust real-time analytics, can lead to intensification of their biomanufacturing along with determination/control product quality. We learn from experience that transforming analytics to implementing process control requires effective data curation, modeling algorithms, and appropriate hardware/software infrastructure supporting biomanufacturing (Ündey et al., 2010). Tailored analytics, automation and increased throughput of critical data entrusts users with the responsibility of maintaining and managing contextualized data

diligently. Frameworks like Industry 4.0/5.0 are available that promote integrating physical components with digital counterparts for improving process efficiency, ensuring consistency, minimize wastes or even anticipate batch failures (Macdonald, 2020). However, ‘smart data utility’ (Wasalathanthri et al., 2020) or hybrid modeling within agile processes and ‘smart manufacturing’ (Gargalo et al., 2020) in general are yet to become an implemented reality for much of biopharma. Within current VV manufacturing, their implementation calls for systems to collect and connect process data, extract meaningful correlations related to process states and indicate product quality. To achieve this, streamlined data collection and effective digestion of ‘big data’ collected using PAT tools or soft sensors is essential (Gargalo et al., 2020). Often, this information is combined with (empirically parameterized) mechanistic models due to increased risk of their lower resolution to disturbances of ineffective capture of time dependencies, whereas such dynamics/non-linearity are common in bioprocesses (Guerra and Glassey, 2018), thus helping streamline intensification or enabling continuous processing (Schmidt et al., 2022).

Challenges in VV manufacturing being met with the incorporation of multivariate data analyses (MVDA) and also machine learning (ML) include those arising from raw material complexity, process variations and continuously evolving analytical tools and also in the identification of higher order CQAs (Emerson et al., 2020; Wilkinson Jr and Irion, 2019).

MVDA has the power to meaningfully digest spectroscopic data due to its ability to inter-correlate information with matrix sample properties and develop summary indices for processes. Depending on the size or amount of data available for modeling exercises versus noise (Silva et al., 2019), the choice of direct analysis using established statistical methods (Kourti and MacGregor, 1995) or dimensionality reduction may be exercised. For example, dimensionality reduction coupled with one-to-many learning methods help discover latent variables and/or patterns or correlations, as seen with the analysis of flow cytometry data to derive cell health trends based on differential fluorescence of cell populations (Cheung et al., 2022). Supervised and unsupervised learning methods respectively can identify pairs of independent and dependent process variables or find hidden or non-obvious data patterns, as shown with upstream operations (Driskell et al., 2008; Goldrick et al., 2020; Konakovsky et al., 2015; Rafferty et al., 2020; Teixeira et al., 2009). Chemometrics – calculations applied to continuous data like time series or multidimensional spectra (Lourenço et al., 2012), is also used to treat Raman, 1D/2D fluorescence data etc., for applications discussed in Section 4.1 (Baradez et al., 2018; Chiappini et al., 2021; Pais et al., 2019; Tulsyan et al., 2020) and have been extensively reviewed (Goldrick et al., 2020; Lourenço et al., 2012; Santos et al., 2018). Reports of chemometrics within multi-attribute monitoring like deconvoluting MS data (Marhuenda-Egea et al., 2013), soft sensing, hybrid learning and biosensor development are also available (Brunner et al., 2021). A list of modeling/data assimilation tools (SIMCA, Unscrambler, OPUS, GRAMS for analysis, SIPAT, xPAT, BioPAT, SynTQ etc.) is available with concurrent developments in soft sensing (process measurements with mathematical modeling) with their use with goals like CPP identification or fault detection from process trajectory predictions (Gerzon et al., 2022; Rathore et al., 2022). Decreased maintenance costs using such software solutions increases the value of manufacturing modalities like VV whose production scales are often associated with high therapy costs (Hosseini and Fath, 2023).

ML in biomanufacturing has previously been used to enable process understanding and control by relating input data (process variables, resultant quality attributes) to output (process efficiency, KPI etc.) with potential extensions to assessing scale-up, computing control limits or diagnose problems in processes (Qiao et al., 2012). Several reviews detail the types of learning applicable to biomanufacturing based on supervision status or reinforcement learning (Helleckes et al., 2023; Rathore et al., 2023), and outline the motivations like real-time control, digital twins and process forecasting etc., through a variety of

methodologies (Walsh et al., 2022; Wilkinson Jr and Irion, 2019). This has helped develop a stepwise framework for employing ML that includes sample selection to facilitate its appropriate partitioning as training, test or validation data (Walsh et al., 2022), pre-processing, selecting independent variables and model building (Rolinger et al., 2020), however, there are no reports of implementation within VV biomanufacturing today. It is also to note that terminologies MVDA and ML have been interchangeably used due to overlap in involved statistical learning (Banner et al., 2021).

Implementation of ML warrants a need for strict model validation due to the data-driven nature of these correlations versus those based on physicochemical phenomena, to ensure no biases from calibration conditions, overfitting or model suitability is present. Standardized and tractable methods for validation of both predictive and analytical models may be the most crucial step to realizing general applicability or use of AI/ML in VV biomanufacturing, as it will provide the necessary confidence and support for regulatory compliance, in which these models are often viewed with nuanced reticence and caution due to the inherently untraceable nature of some advanced models and their predicted correlations. A judicious model validation pipeline hence first requires the partitioning of data into distinct sets for training, testing, and validation. The training set, representing a substantial portion of the available data, facilitates the initial learning phase of the model, allowing it to capture patterns and relationships internal to or amongst the unit operations, CQAs, and CPPs. Acquiring representative and diverse training data that accurately reflects the complexity and variability of bioprocessing conditions and outcomes can be challenging. Collaborating with domain experts and incorporating data from various bioprocessing scenarios can enhance the representativeness of the training data. Ensuring that the training set covers a comprehensive spectrum of conditions, including different cell lines, culture media, and operating parameters, can mitigate biases and improve model robustness. The isolated testing set serves as a critical benchmark, assessing how well the model generalizes to new, unseen data—a vital consideration in the dynamic and variable environment of VV production. A random selection of samples for the testing set and careful consideration of the distribution of key parameters can help maintain the integrity of the testing phase. Lastly, the validation set, distinct from both training and testing data, becomes instrumental in fine-tuning the model and ensuring its adaptability to real-world scenarios encountered in bioprocessing operations. But as more data is utilized in training and testing sets, suitable validation data can be harder to curate and mine – which at this stage, may also exacerbate challenges in data quality.

Meticulous data partitioning and model validation is essential to circumvent automatic data stratification, and the efficacy of applied approaches can be quantitatively assessed through mathematical parameters derived from regression analysis, e.g. root mean square error. Two overarching validation approaches, data-oriented and hypothesis-oriented, involve empirical verification of correlations and reliance on first-principles theory, respectively. Within each of these categories, statistical techniques like ANOVA or bootstrap cross validation or component analyses with further contextualization using process information may be performed (Westad and Marini, 2015). Furthermore, to ensure wide applicability of these models and prevent inherent bias (Ho et al., 2020), external validation (using independent data) of these models may also be required, especially in medicine and related fields (Cabitza et al., 2021). While reports of such validations have not quite been translated to VV biomanufacturing, their potential significance remains underscored.

This said, across the bioprocessing universe, there are potentially applicable or translatable validation techniques for feature checking – both categorical (e.g., data type, quality, outliers) and numerical—such as coefficients of determination, maximum likelihood, goodness of fit, and information criteria (Rathore et al., 2023), have been proposed. Considerations of appropriate scaling and weighting of input variables are critical, capturing diverse process variations within calibration

datasets (Galinha and Crespo, 2021). Additionally, models for consolidating process knowledge (Helleckes et al., 2023) and evaluating performance (Wasalathanthri et al., 2020) have been identified, underscoring the ongoing efforts to advance the reliability and applicability of data-driven models that may soon find a major role in VV biomanufacturing.

Using just UV, pH and conductivity data, (Nikita et al., 2022) demonstrated guidelines to select the best ML model amongst five tested that included deep NN and support vector regression (SVR), to predict CQA like concentration, charge variants in mAb purification. (Scharl et al., 2023) implemented ANNs with chromatographic separations where they utilized FTIR spectra for its training with product and impurity concentration information and studied points of sensor failure and optimizing feature selection. SVR was also used with Raman data to model and predict cell culture parameters in real time (Tulsyan et al., 2020). (Emerson et al., 2020) recommend unsupervised learning methods for best process understanding, especially in this domain due to issues with *completeness* of datasets, which must be actively pursued to bridge the gap within VV manufacturing. Another realistic problem with multiple PAT sensors feeding learning models is appropriate data fusion, which has been reported, with suggestions for resolution (Rolinger et al., 2020). In the future, transfer learning (TLP) type ML may help address data insufficiency issues by utilizing abundant datasets of closely related domains, as also indicated by (Oyetunde et al., 2018). ML to study virus structure/biochemistry, predict vector genotoxicity, design stable capsids, create new variants, and for data analysis like infectivity prediction using microscopic images or linking pH and metabolic activity have been reported (Bryant et al., 2021; Dodkins et al., 2023; Schwarzer et al., 2021; Wec et al., 2021; Williams et al., 2023; ZHU et al., 2021).

While there is an evident drive for the need for PAT in place to ensure availability of complete datasets for better training purposes (Rathore et al., 2023), their presence may also eventually lead to the 3'V' challenge (volume, variety and velocity of data) (Bisschops and Cameron, 2022) – strategies to solve which can also be their own field of opportunities. Validation challenges are also likely, due to the current state of developing, non-standardized methods and a range of properties like sample preparations, costs and fits-for-purpose (Sarkar and Lin-Gibson, 2021). It is hence important for us to ensure PAT development and learning/data management sync to ensure timely to ensure timely evolution and implementation. The FDA has recently released discussions elucidating the regulators perspective with adopting advanced data science or ML within biomanufacturing (FDA, 2023a) along with other areas of pharmaceutical development and a list of helpful related guidelines (FDA, 2023b), that can be used for structure within the development space.

6. Outlook and opportunities

While the breadth and physicochemical complexity of VV challenges broad applicability of specific analytics, it is still vital to investigate inline/online/at-line characterization facilitators for discerning CQAs in a rapid and reliable manner, with high detection sensitivity and easy application or analysis. In this review, we discussed the current state of analytics for characterizing VV products with a critical perspective on their applicability as real-time enabling PAT. In this process, we additionally presented both physical and analytical opportunities within each 'line' category (Fig. 3) that can help accelerate this process.

The future of real-time analytics can be (bio)sensors and similar advanced electronics, wireless technology, high power computing for rapid analysis and process control, and conduits that can directly be connected to process streams at scale. Automated systems for timely sample collection while enabling at-line measurements already exist (e.g., Fisp® probes, MAST, CEDEX Bio HT, SegFlow, PATROL, Flow Injection Analyzers etc., (Wasalathanthri et al., 2020)) and must be continuously validated. While certain analyses might be rate-limited due to biochemical interactions involved, innovation in materials and

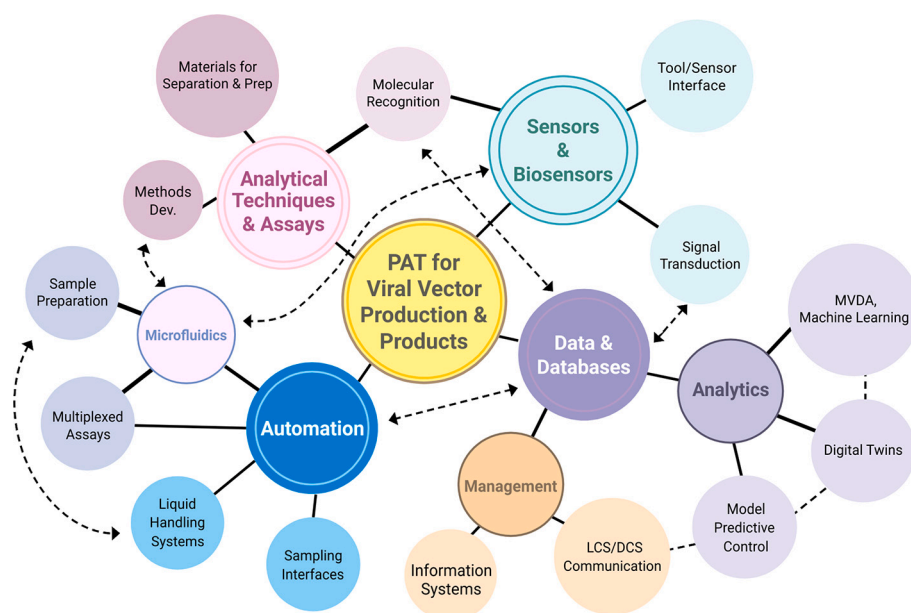


Fig. 6. Outlook and potential opportunities for scientists and engineers across disciplines in developing PAT for cell and gene therapies in general. Several areas of overlap between PAT-enabling analytical development, automation strategies, sensor design and PAT-dependent data curation, analysis and insight derivations can be visualized through the overlaps (represented with interconnections and dashed lines), calling for increased multidisciplinary collaborations.

methods for efficient sample preparation or multiplexing via microfluidics are avenues for development. High throughput methods with higher readout precision compared to manual (Christler et al., 2020) have also seen success but may not always be process compatible (Silva et al., 2021). Critical comparisons of liquid handling systems and microfluidics for parallelization is available, where researchers emphasized on the latter for throughput and separations, especially with smaller sample volumes and at point of care (Anderson et al., 2022; Christler et al., 2020; Layton et al., 2013; Moutsatsou et al., 2019; Wasalathanthri et al., 2021). Integrating microfluidic systems that enable nanoliter scale volume handling capacities with analytical technologies that have lower sensitivities or higher limits of detection, may also abet with improving these ranges within their instrumentation, as seen in several cases (Kakaç et al., 2009; Maddali et al., 2021; Ou et al., 2020; Pinto et al., 2021; Prodromidis and Economou, 2017; Silva et al., 2021; Zhang et al., 2018), taking limits of detection to femtomolar ranges (Ozer et al., 2020).

Continued advancements in aspects of materials research for improved sample capture, separation or signal attenuation is another area of focus important for efficient PAT development, as seen with the design of bio-nanoparticles for virus detection (Yur et al., 2021), creative use of carbon nanotubes within portable microfluidics, non-invasive virus detection and chip reusability (Yeh et al., 2020), or novel polyvinyl alcohol films for capillary driven, rapid, consistent sample preparation for TEM (Hauser et al., 2020). Analyses using low sample volumes can be challenging due to need for enrichment, which was also solved through micropatterned Si nano-vial incorporation (Ekström et al., 2001). Helpful reviews with information on microfluidic fabrication, immobilization, extraction and detection methods to use them for sample preparation and analysis (Alidoust et al., 2021; Sonker et al., 2017) have been authored. These ideas are waiting for implementation within novel PAT development, and it may be beneficial to compare them against existing technologies in terms of sample throughput, instrument and disposable costs, reagents, number of output parameters and output accuracy, or a mnemonic “5 M” – material, mechanisms (of interaction or detection), modalities (to accommodate a variety of vectors), multiplexing (multiple assays in one) and multiple use (for best economics).

The design and adaptation of multi-attribute methods (MAM) also holds importance as the possible future of analytics landscape for both

mAb and VV, with reported success with the former (Rogers et al., 2018) due to their powers of simultaneous attribute monitoring, even complex like PTMs (Zhang and Guo, 2017). Interesting methods including multi-loop sample transfers, multi-port fraction collection, automated column switching, fractionation such as ‘heart-cutting’ (Song et al., 2021; Williams et al., 2017), or diversion to waste have enabled their automation and integration at-line. Reports of 2D HPLC coupled with an ESI-MS for effective separation and capsid profiling, heterogeneity analysis and PTM identification of AAVs (Wu et al., 2022b), and IEX-HPLC coupled with SEC-MALS for particle information plus AAV product heterogeneity (Gagnon et al., 2021) are examples of MAMs for VV. Given lability and stability concerns of VVs, such MAMs that adhere to quality by design through molecular level inquiry of the sample (Zhang and Guo, 2017) may be the future of quality control in their biomanufacturing, ultimately supporting batch release (Camperi et al., 2021).

As a closing note, we would like to resonate with Eisen et al., 2020 and reemphasize that “technology change requires mindset change”. Through Fig. 6, we summarize future opportunities in PAT development into four broad categories. While we document recent developments in analytics and biosensors for VV, making them true PAT requires crossing the barrier from lab into development or manufacturing environments, at scale. Scientists and engineers in this space should thus consider increasing throughput, automation, and effective data management into their development designs, which also ultimately potentiates process control. As seen in Fig. 6, developing PAT for VV is a truly interdisciplinary endeavor with the power to assimilate chemical, biomedical, electronics, electrical, computer, mechanical and data engineers, together with physicists, chemists, biologists on the same projects, with strong entrepreneurial opportunities as well. Academic – industry partnerships and collaborations can enable and accelerate these efforts to success, with mutual benefits of learning, synchronous development and credibly inform the evolving regulatory policies on developing process analytical technologies.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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